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Effects of perceived environmental quality and psychological status on outdoor thermal comfort: a panel study in Southern China

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Effects of perceived environmental quality and psychological status on outdoor thermal comfort: a panel study in Southern China

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Abstract

Urban residents are exposed to multiple environmental stimuli. Past studies have examined the influence of heat stress on outdoor thermal comfort, but the cross-modal effects of various sensory stimuli on outdoor thermal comfort remain unclear. This study used chi-square tests and logistic regression to examine the cross-modal effects of perceived environmental quality (acoustic, visual and air quality) and psychological status (irritability, perceived tiredness) on outdoor thermal comfort. We conducted thermal walk experiments in Guangzhou, China, in September and October 2022. This panel study consisted of mobile environmental sensors and thermal comfort surveys by fourteen university students on three walking routes, representing urban renewal sites, new urban areas, and urban blue-green spaces. The 'very bright' (OR=15.76, p<0.001) and 'glaring' groups (OR=4.09, p<0.001) were likelier to feel hotter than the 'dark' and 'not glaring' groups. People who found the noise level annoying were likelier to feel uncomfortable (OR=1.61, p=0.004). Moreover, people perceiving air quality as 'very good' were less likely to find the thermal environment unacceptable (OR=0.27, p=0.035). The irritated (OR=4.18, p<0.001) and tired groups (OR=1.81, p=0.002) were likelier to find the thermal environment intolerable. Our findings highlight the need to consider multisensory comfort and psychological status to inform sustainable urban design.

Keywords: multisensory interaction, perceived environmental quality, outdoor thermal comfort, acoustic comfort, visual comfort, perceived air quality, urban environment

Highlights

- Multisensory comfort is assessed by mobile measurements and thermal walks.
- Sun and glare sensation vote influence thermal comfort.
- Noise annoyance vote has a cross-modal effect on thermal comfort.
- Thermal comfort is influenced by air quality vote.
- Irritability and perceived tiredness affect thermal comfort.

Nomenclature

ADV	Air dustiness vote
AQV	Air quality vote
BLR	Binary logistic regression
BMI	Body mass index
CO2	Carbon dioxide
GSV	Glare sensation vote
HSV	Humidity sensation vote
LCZ	Local climate zone
MEMI	Munich Energy-balance Model for Individuals
NAV	Noise annoyance vote
OLR	Ordinal logistic regression
PANAS	Positive and Negative Affective Schedule
PET	Physiological Equivalent Temperature (°C)
POMS	Profile of Mood States
PM	Particulate matter
RCP	Representative Concentration Pathway
RH	Relative humidity (%)
SSP	Shared Socioeconomic Pathway
SSV	Sun sensation vote
SVF	Sky view factor
T_a	Air temperature (°C)
T_g	Globe temperature (°C)
T_{mrt}	Mean radiant temperature (°C)
TAV	Thermal acceptance vote
TCV	Thermal comfort vote
TSV	Thermal sensation vote
TTV	Thermal tolerance vote
v	Wind speed (m/s)
WSV	Wind sensation vote
φ_c	Cramer's V

1 Introduction

In assessing urban overheating risks, increasing heatwaves associated with climate change and urban heat islands increase the risk of heat exposure for urban populations (Nazarian et al., 2022; Tong et al., 2021; Wang et al., 2021). According to most Shared Socioeconomic Pathway (SSP) scenarios (SSP2.4-8.5), East Asia will likely experience an increase of > 2 °C by the end of the 21st century (You et al., 2022), which will result in serious health concerns (Zhang et al., 2023). On average, globally, urban heat island warming will probably be equivalent to about half the warming caused by climate change by the year 2050 (Huang et al., 2019). For instance, in a city that experiences 2 °C of warming from climate change, that would mean an extra degree of warming. Due to past warming trends, globally there were 7 heat-related excess deaths per 100,000 residents in 2000–2019 (Zhao et al., 2021). In China, the attributable deaths to heatwaves have increased fourfold from 3679 in the 1980s to 15,500 in the 2010s (Chen et al., 2022c). The proportion of attributable deaths to heatwaves for people with cardiovascular disease increased from 42.7% in the 1990s to 59.4% in the 2010s in China (Chen et al., 2022c). Moreover, the death toll caused by heatwaves in 2090 will increase under Representative Concentration Pathway (RCP)2.6 (20,303 deaths), RCP4.5 (35,025 deaths) and RCP8.5 (72,260 deaths) (Chen et al., 2022b). Approximately half of all heatwave-related deaths will occur in east and central China under any scenario, encompassing developed urban areas in China (Chen et al., 2022b). This issue is particularly urgent for vulnerable populations such as the elderly (Cai et al., 2022) and outdoor workers (Ebi et al., 2021).

In China, urban warming accounted for about a third (0.49 °C) towards the overall warming trend in annual mean temperature from 1961–2013 (1.44 °C) (Sun et al., 2016). With a significant

contribution of urban heating in cities towards overall worldwide warming (Hamdi et al., 2020), healthy city design has been proposed to alleviate the increasing heat stress in cities (Frumkin et al., 2020). Sustainable urban design at the neighbourhood and building scales is crucial to enhance the outdoor thermal environment, thus keeping the city cooler and improving the comfort of urban residents (Jay et al., 2021).

People in the outdoor environment are influenced by multisensory stimulants, which interact with each other. Previous studies have focused on the unimodal effects of visual factors on outdoor visual comfort (Achsani & Wonorahardjo, 2020; Compagnon & Goyette-Pernot, 2004), noise on acoustic comfort (Kang et al., 2016) and air pollution levels on air quality perception (Pantavou et al., 2017; Pantavou et al., 2018). Other studies have indicated a cross-modal effect of visual comfort (Nitidara et al., 2022), acoustic comfort (Lau & Choi, 2021; Ma et al., 2021) and perceived air quality (Cureau et al., 2022) on thermal comfort. Yet, the combined impact of multi-domain comfort (visual, acoustic and air quality) on thermal comfort is not well understood. To investigate the effects of personalised characteristics of heat exposure, it is necessary to use portable sensing devices, multi-domain mobile transects (Kousis et al., 2022) and surveys to consider the interaction between perceived environmental quality factors (visual, acoustic and air quality) and thermal factors. Personal heat and environmental exposure could differ substantially from those at the population level (Tuholske et al., 2021), which requires the use of portable sensors in different urban locations (Bailey et al., 2020). In particular, personal environmental exposure differs substantially due to differences in urban morphology (Dzyuban et al., 2022; Pigliautile & Pisello, 2020; Schnell et al., 2021), human factors (e.g., behavioural patterns, physiological and psychological responses) (Nazarian & Lee, 2021) and personal activities (Chaix, 2018). Apart from the interaction between multi-domain comfort, Schreuder et al. (2016) highlighted the lack

of empirical evidence on the interaction between multisensory comfort and emotional responses. This study contributes to the field in the following ways:

- We develop survey questions to comprehensively assess multi-domain comfort in outdoor settings, which can be applied in cities in various climates.
- We analyse multi-domain comfort and psychological status using mobile transects, which provide a better understanding of the impact of multisensory interaction and psychological factors on outdoor thermal comfort in transient settings.

Fig. 1 shows the theoretical framework of this study **(**Gordon & Leon, 2005**;** Kuras et al., 2017**;** Liu et al., 2023a**;** Ma et al., 2021**)**. This framework highlights the role of personal environmental exposure and outdoor multisensory interactions as contributing factors to outdoor thermal comfort. Moreover, we acknowledge thermal comfort as an early warning signal to prevent more serious heat-related illnesses from developing, which has implications for public health.

In short, pedestrians are exposed to multiple stimulants in outdoor environments, but the interaction between various outdoor environmental factors and its effect on thermal perceptions remain unclear. Besides multisensory interaction, the effects of psychological status on outdoor thermal comfort are not clearly understood. To address these issues, this study aimed to examine the cross-modal effects of perceived environmental quality (visual comfort, acoustic comfort, and air quality) and psychological status on outdoor thermal comfort. Specifically, we answer the following research questions.

1) What is the relationship between perceived environmental quality (visual comfort, acoustic comfort, air quality) and outdoor thermal comfort?

2) What is the relationship between psychological status and outdoor thermal comfort?

3) To what extent do visual comfort, acoustic comfort, perceived air quality and psychological status predict people's perception of outdoor thermal conditions?

To answer these questions, we conducted mobile transect measurements, including a mobile meteorological station on a backpack and a concurrent outdoor environmental quality survey. This human-centric approach quantifies the effects of personalized characteristics of environmental exposure on human health at a finer spatial scale (Nazarian & Lee, 2021). Three locations in Guangzhou, China, were selected to represent different landscape characteristics: Yongqingfang, a typical old urban block; Knowledge City, a new urban development site; Haizhu Wetland Park, a typical urban greenspace.

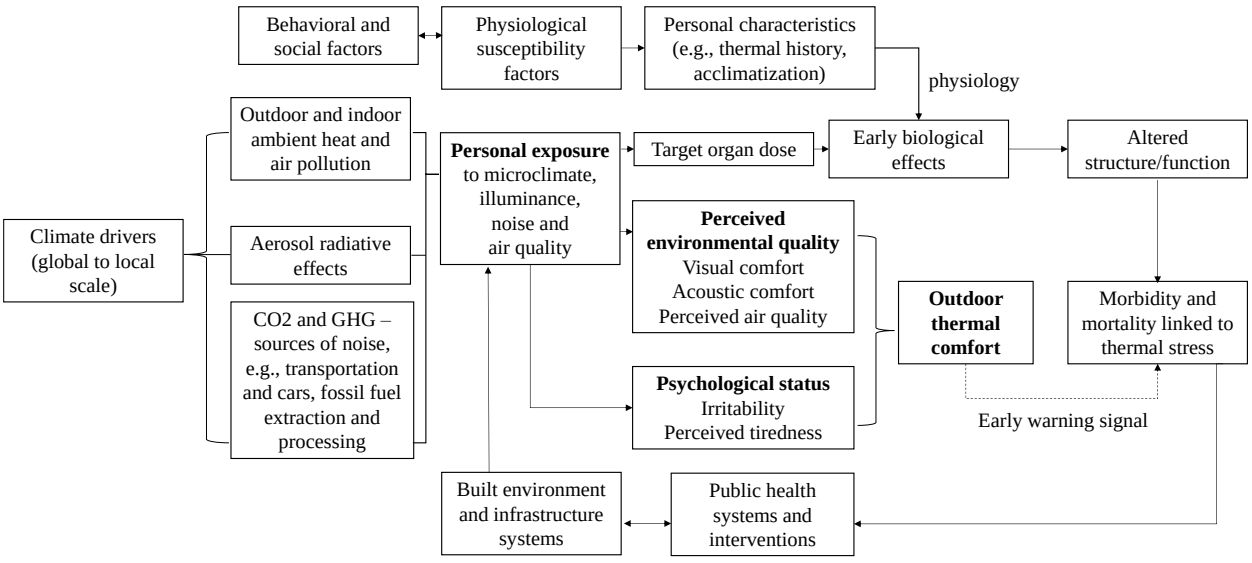


Fig. 1 Theoretical framework linking climate drivers, personal environmental exposure and multisensory perception, with outdoor thermal comfort as an early warning signal for adverse health events linked to thermal stress. This framework is adapted from the following sources (Gordon & Leon, 2005; Kuras et al., 2017; Liu et al., 2023a; Ma et al., 2021).

2 Materials and methods

2.1 The study area and regional climate

Guangzhou is located on the southeast coast of China (23°7’ N 113°15’ E) and experiences a humid subtropical climate (Köppen Cfa) (Fig. 2). From 1981 to 2010, the mean maximum temperature and average relative humidity in September in Guangzhou is 32.1 °C and 72%, respectively (China Meteorological Administration, 2012). The mean September wind speed in Guangzhou is 1.5 m/s (China Meteorological Administration, 2012). Moreover, the mean total daily global radiation is 166.5 W/m^2 for Guangzhou between 2000 and 2010 (World Radiation Data Centre, 2019). Summer in Guangzhou is featured by high temperature and humidity associated with the region’s monsoons.



Fig. 2 Location of the study area. (a) Map of China, (b) Location of Guangzhou, (c) Haizhu Wetland Park, (d) Knowledge City, (e) Yongqingfang.

Guangzhou is one of the pilot cities of the World Bank's sustainable urban cooling project in China. In this project, we selected three sites for our pilot study: Yongqingfang, Knowledge City, and Haizhu Wetland Park (Fig. 2). We classified our sites according to the local climate zone (LCZ), which is an urban surface classification system considering urban built forms and land cover types (Stewart & Oke, 2012). Yongqingfang (LCZ 3 – compact low-rise) is an urban renewal site located about 8 km from the city centre, with historical buildings renovated in 2016. It has a dense mix of low-rise buildings (1-3 stories) and narrow streets with limited urban vegetation. Knowledge City (LCZ 4 – open high-rise) is a newly developed urban area built in 2010, which is about 30 km away from the city centre. It features an open arrangement of tall buildings (mean building height of 40 m), wide streets and scattered vegetation. Haizhu Wetland Park (LCZ A – dense trees) is an urban wetland (1,100 hectares) about 6 km from the city centre, with abundant trees on mostly pervious land cover, a delta city lake and a river wetland. These study locations represent old urban blocks, new urban development sites and typical urban greenspace.

2.2 Experimental setup

2.2.1 Thermal walk experiments and survey campaigns

The thermal walk survey campaign was conducted in Knowledge City (25/9/2021), Haizhu Wetland Park (27/9/2021), and Yongqingfang (clear day: 29/9/2021 and cloudy day: 20/10/2021). We conducted the thermal walk on each experiment day at the following times: 09:30-11:00, 13:30-15:00, and 16:30-18:00, representing the diurnal cycle of the thermal environment. Three walking routes were designed for Yongqingfang, Knowledge City and Haizhu Wetland Park (Fig. 3). The route at Yongqingfang passed through the Cantonese Opera Museum, a small river, arcade streets and alleys, with 16 survey points. The route at Knowledge City consisted of a commercial

district, a path by a river, a street sidewalk and a science park, with 14 survey points. The route at Haizhu Wetland Park passed through an exposed square, pathways near a lake, forest mosaic composite wetlands, pavilions, and lawn areas, with 10 survey points. Each walking route was about 1 km long and took about 60 minutes.

To study people's perceptions under different environmental conditions and obtain more comprehensive results, we chose survey points with significant changes in the surrounding landscape, such as corners of streets and exits of alleys. Therefore, more survey points were selected in the old urban block Yongqingfang and the new development block Knowledge City owing to high landscape heterogeneity. In contrast, fewer survey points were chosen in the Haizhu Wetland Park due to relatively low landscape heterogeneity.



Fig. 3 Thermal walk experiment routes, showing a) Yongqingfang, b) Knowledge City, and c) Haizhu Wetland Park. Fixed weather stations were shown as stars at Yongqingfang and Haizhu Wetland Park, where a fixed weather station was set up at point 1 in Knowledge City.

We conducted a panel study and used a cluster sampling method to recruit 14 college students from Guangzhou in the thermal walk experiment. All 14 participants took part in three thermal walks on each experiment day (09:30-11:00, 13:30-15:00, and 16:30-18:00). The thermal walk lasted 1 to 1.5 hours, depending on the walking routes. In Yongqingfang, the sample size was 14 participants×16 survey points×3 rounds×2 days = 1,344. In Knowledge City, the sample size was

14 particpants×14 survey points×3 rounds = 588, whereas the sample size in Haizhu Wetland Park was 14 participants×10 survey points×3 rounds = 420. A total of 2,352 survey responses were included in our analysis. An a priori power analysis was conducted using G*Power version 3.1.9.7 (Faul et al., 2009) to determine the minimum sample size required to test the study hypothesis. Results indicated the required sample size to achieve 80% power for detecting a medium effect, at a significance criterion of $\alpha = .05$, was $n = 193$ ($df = 12$) for a chi-square test of association. For logistic regression, the minimum sample was calculated using the formula $n = 100 + 50i$, where i is the number of independent variables in the model (Bujang et al., 2018). Our logistic regression model has 13 independent variables, so the minimum required sample size would be 100 + 50×13 = 750. Overall, the obtained sample size of $n = 2,352$ is adequate to test the study hypothesis.

We acknowledge heat storage from direct sunlight and physical activity during the experiments may affect participants' thermal sensations (Xu et al., 2021). To account for the impact of heat storage on thermal sensation, we scheduled rest periods between the three experiments conducted in a day. Participants rested in air-conditioned mall shops in Knowledge City and Yongqingfang, as well as in outdoor shops with shade and parasols in Haizhu Wetland Park. Additionally, we arranged for survey campaigns to be conducted on alternate days, allowing participants a full day of rest after each survey campaign.

A baseline survey was given to each participant to understand their demographic background, such as gender, age, height, weight, place of origin, and length of stay in Guangzhou. We asked the participants to wear the same summer clothing (a light-coloured short-sleeved shirt, long pants, a cap, and a pair of socks and shoes) to keep the clothing insulation value (clo) at 0.51. The clothing insulation value was derived from ISO 9920 (2007) (Table S1).

Table 1 shows the characteristics of our study participants, which involved 14 subjects (7 males and 7 females) acclimatized to the local climate in Guangzhou (the capital city of the Guangdong Province). The mean age of participants was 20.2 ± 0.4 as they were college students. Most of our participants were from the Guangdong Province (92.9%), and over 71.4% of participants had stayed in Guangzhou for more than one year. For those participants who have lived in Guangzhou for less than three months, they were originally from other cities in the Guangdong Province, which have a climate similar to Guangzhou. Therefore, our study participants were acclimatized to the local climate.

Table 1 Demographics of the study participants (*n* = 14)

Variables	*n* (%) or mean ± SD
Age (y)	20.2 ± 0.4
Sex (male)	7 (50%)
Body Mass Index (BMI) (kg/m^2)	21.3 ± 1.9
Place of origin	
Guangdong Province	13 (92.9%)
Gansu Province	1 (7.1%)
Length of stay in Guangzhou	
<3 months	4 (28.6%)
3-12 months	0 (0%)
1-3 years	8 (57.1%)
3-10 years	0 (0%)
>10 years	2 (14.3%)

At each survey point, the participants were asked to fill out the survey questions regarding their perceived environmental quality, including the thermal, visual, acoustic, and air quality environments (ISO 10551, 2019), as well as their psychological status (irritability and perceived tiredness) (Lam et al., 2021b; Liu et al., 2017). At each stop, we asked the participants to stand for

over three minutes before responding to the survey, in order to account for the time-lag effect of meteorological parameters on thermal sensation (Lau et al., 2019). Our supplementary material includes the survey questions for the baseline survey and thermal walk experiment.

The survey questions on the thermal environment were designed according to ISO 10551 (2019), including thermal sensation vote (TSV), thermal preference, thermal comfort vote (TCV), thermal acceptance vote (TAV) and thermal tolerance vote (TTV). We extended the ASHRAE 7-point to 11-point scale to account for our participants' exposure to the hot and humid outdoor environment in Guangzhou. The 11-point scale was introduced in the thermal sensation index model (TSENS) (Gagge et al., 1986; Soligo et al., 1998) and later adopted in studies in various climate zones (Chen et al., 2018; Khatun et al., 2017; Yin et al., 2021). In particular, we added +4 (very hot) and +5 (extremely hot) to capture people's thermal sensations in hot and humid conditions in Guangzhou, which was shown to be valid in a previous Guangzhou study (Fang et al., 2019b). We removed -4 (very cold) and -5 (extremely cold) from our analysis as these conditions were not encountered in this study, resulting in a 9-point scale (-3: cold to +5: extremely hot) like Fang et al. (2019b). As Guangzhou experiences a hot and humid summer, we added a five-point scale of humidity sensation vote (HSV) and wind sensation vote (WSV), which have been used in previous studies (Heng & Chow, 2019; Lam et al., 2021a).

The questions for assessing the visual environment were designed based on ISO 10551 (2019). We included a question on whether the participants were under shade to understand the impact of shade on visual perception. Apart from the sun sensation vote (SSV), we also included a question on the glare sensation vote (GSV), which was modified from Iwata and Tokura (1998). The questions for assessing the acoustic environment were designed based on ISO 10551 (2019). We

included a question on noise annoyance vote (NAV) for the noise level. Moreover, we included a question on the sound types (Jeon & Jo, 2020) to capture sounds perceived as pleasant or unpleasant by people (Chen et al., 2022a). The questions for assessing outdoor air quality were divided into two parts. The first part is about olfactory comfort (i.e., the smelliness level) perceived by people (ISO 10551, 2019), whereas the second part concerns the perceived air dustiness vote (ADV) and air quality vote (AQV) (Pantavou et al., 2017). Moreover, we include questions on respondents' irritability and perceived tiredness to indicate their psychological status during the thermal walk experiments. Overall, these survey questions present a comprehensive approach to understand people's perception of outdoor environmental quality.

2.2.2 Mobile monitoring system

We developed a mobile environmental system that can be worn on a backpack for the thermal walk experiments (Fig. 4). The instruments were carried by other volunteers who did not participate in the survey in this study. The survey participants did not carry the instruments. Table 2 shows the specifications of the instruments used in this study. They consisted of a Kestrel 5400 Heat Stress Tracker that measured air temperature (T_a), relative humidity (RH), wind speed (v) and globe temperature (T_g), a TES 1339R lux meter that measured the horizontal illuminance, an Aicevoos AS-K8 sound level meter for measuring decibel, and an HR-500 portable air quality sensor for measuring outdoor air quality (i.e., particulate matter – PM1, PM2.5, PM10, and carbon dioxide – CO2).

The mean radiant temperature (T_{mrt}) was calculated using Eq. (1)(ASHRAE, 2001):

$$T_{mrt} = \left[\left(T_g + 273.15\right)^4 + \frac{1.10*10^8*v^{0.6}}{\varepsilon*\mathrm{D}^{0.4}}\left(T_g - T_a\right)\right]^{\frac{1}{4}} - 273.15 \quad (1)$$

where T_{mrt} represents the mean radiation temperature, T_g is the black-globe temperature (°C), T_a is the air temperature (°C), v represents the wind speed (m/s), D is the diameter of the black globe (0.025 m), and ε is the globe emissivity (0.95 for the black globe).

Table 2 Specifications of the environmental monitoring equipment used in this study.

Measured parameter	Logger	Measurement range	Accuracy	Measured interval
Air temperature (°C)	Kestrel 5400 Heat Stress Tracker	-29 °C – 70 °C	±0.5 °C	2 s
Relative humidity (%)		0 – 100%	±2%	2 s
Wind speed (m/s)		0.6 – 40 m/s	±3%	2 s
Globe temperature (°C)		-29 °C – 60 °C	±1.4 °C	2 s
Illuminance (lx)	TES 1339R lux meter	0.01 lx – 999,900 lx	±3% of readings	1 s
Decibel (dB)	Aicevoos AS-K8 sound level meter	30 – 130 dB	±1.5 dB	1 s
PM1 (μg/m^3)	HR-500 portable air quality sensor	0 - 500 μg/m^3	0.1 μg/m^3	1 s
PM2.5 (μg/m^3)		0 - 500 μg/m^3	0.1 μg/m^3	1 s
PM10 (μg/m^3)		0 - 500 μg/m^3	0.1 μg/m^3	1 s
CO2 (ppmv)		400 - 2000 ppmv	1 ppmv	1 s

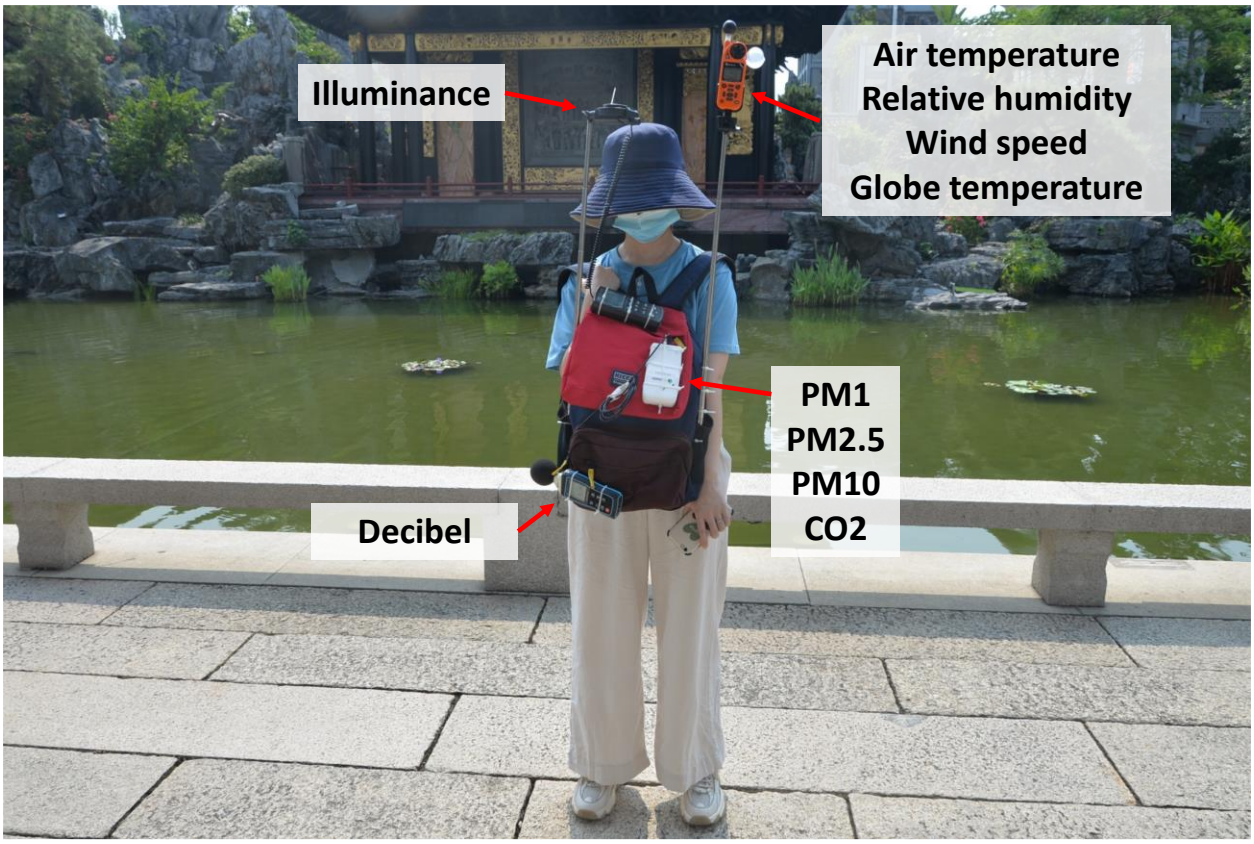


Fig. 4 Mobile environmental monitoring equipment, showing Kestrel 5400 Heat Stress Meter, TES 1339R lux meter, Aicevoos AS-K8 sound level meter, and HR-500 portable air quality sensor.

We chose the Physiological Equivalent Temperature (PET) as our thermal indices as it has been used in our previous study (Deng et al., 2023) and is also widely adopted in outdoor studies (Potchter et al., 2018, 2022). PET is defined as 'the air temperature at which, in a typical indoor setting, the heat budget of the human body is balanced with the same core and skin temperature as under the complex outdoor conditions to be assessed' (Höppe, 1999, p.71). For PET, the standardized person is featured by a metabolic rate of 80 W on top of basic metabolism, and a heat resistance of clothing of 0.9 clo (Höppe, 1999). The PET used the Munich Energy-balance Model for Individuals (MEMI) to present the human heat budget (Matzarakis & Mayer, 1996). We used Rayman Pro 2.1 to calculate the PET (Matzarakis et al., 2010). The input for calculating the PET

was T_a, RH, v and T_{mrt}, together with a clothing insulation of 0.51 clo and metabolic rate of 125 W (at rest, standing) (ISO 8996, 2004).

2.3 Statistical analysis

We conducted a chi-square test of association to determine whether a significant association exists between perceived outdoor environmental quality indicators (i.e., visual, acoustic, and air quality environments) and thermal comfort indicators (TSV, TCV, TAV, and TTV). A chi-square test of association was also conducted to determine if there was an association between psychological status (i.e., irritability and perceived tiredness) and outdoor thermal comfort (research question 2). We chose Cramer's V (ϕ_c) as the effect size statistics as our chi-square test tables are greater than 2 by 2. The ϕ_c indicates the association between two variables, ranging from 0 (no association) to +1 (perfect association). In particular, the value of ϕ_c indicates a small (0.1), medium (0.3) or large effect (0.5). Then, we used a Bonferroni-corrected z-test to indicate which pairs of columns for a given row in different paired groups (perceived outdoor environmental quality indicators/psychological status and outdoor thermal comfort) have statistically significantly different proportions.

We used binary logistic regression (BLR) and ordinal logistic regression analyses (OLR) to investigate which indicators of outdoor environmental quality (perceived visual comfort, acoustic comfort, and air quality) and psychological status significantly predict our participants' outdoor thermal comfort. The BLR examined which indicators of perceived outdoor environmental quality and psychological status significantly predict people's thermal sensation and acceptance. The dependent variable is a binary variable (TSV: -1 to +1 - slightly cool to slightly warm vs +2 to +5: warm to extremely hot, and TAV: acceptable and unacceptable). We omitted TSV responses

between -3 (cold) and -2 (cool), because we wanted to compare TSV between the thermal neutral range (TSV: -1 to +1) and people who felt warm and above (TSV: +2 to +5). The independent variables are the perceived outdoor environmental quality indicators (e.g., SSV, GSV, NAV, Smelliness, ADV and AQV) and psychological status (irritability and perceived tiredness). We adopted BLR for TSV instead of OLR since the TSV model violates the proportional odds assumption. The proportional odds assumption assumes that the odds ratio is the same regardless of where the outcome categories are dichotomized (Kleinbaum & Klein, 2002). Violating this assumption could cause invalid results, so we used a separate binary logistic regression (BLR) model instead (Bender & Grouven, 1998). Moreover, reducing TSV into a smaller number of categories has been adopted in previous research that used logistic regression (e.g., TSV: -1 to +1 and TSV > +1) (Albadra et al., 2017; Lam et al., 2021b). Furthermore, we conducted OLR analyses to determine whether our participants' TCV and TTV could be predicted from perceived outdoor environmental quality indicators and psychological status. The ordinal dependent variable is TCV and TTV (with three or more levels), whereas outdoor environmental quality indicators and psychological status are our dependent variables. The OLR model has been used in previous research assessing outdoor thermal environments (Ali & Patnaik, 2018; Lai et al., 2018). We considered an alpha value of 0.05 as significant, and SPSS version 20 was used for all data analyses.

3 Results

3.1 Observed environmental conditions and site characteristics

Fig. 5 shows the range of meteorological parameter variations for different experimental sites and weather. On a clear day, the median temperature T_a was 32.86 °C, 33.10 °C and 34.08 °C at Knowledge City, Haizhu Wetland Park and Yongqingfang, respectively (Fig. 5a). For clear day

conditions, Haizhu Wetland Park had a higher RH (65.77%) than Knowledge City (60.69%) and Yongqingfang (57.07%) (Fig. 5b), because Haizhu Wetland Park has high vegetation cover and potentially higher evapotranspiration.

The median horizontal illuminance at Knowledge City was higher (12,358 Lx) than at Haizhu Wetland Park (7,597.58 Lx) and Yongqingfang (4,590.81 Lx) (Fig. 5c). Moreover, the distribution for each period on each route is shown in the following figure. Moreover, the distribution for each period on each route is shown in the following figure. Overall, the illuminance was highest between 9:30-11:00 and lowest between 16:30-18:00 (Fig. A1 in Appendix A). However, the median PET of 36.52°C in the Knowledge City was only 0.64°C and 0.37°C higher than Haizhu Wetland Park and Yongqingfang, respectively.

The median sound intensity at Haizhu Wetland Park (54.83 dB) was lower than Knowledge City (61.29 dB) and Yongqingfang (62.56 dB) (Fig. 5d). This result is mostly due to the natural sound types at Haizhu Wetland Park compared with traffic and construction noise at Knowledge City and Yongqingfang.

Among the three sites, Knowledge City has the lowest pollutant concentrations (Fig. 5e). For clear conditions, the median PM2.5 concentration in Knowledge City was 7.80 μg/m³ compared with Haizhu Wetland Park (46.92 μg/m³) and Yongqingfang (50.44 μg/m³) The median CO2 concentrations were also lowest in Knowledge City (505.22 ppmv) and highest in Yongqingfang (668.73 ppmv) (Fig. 5f).

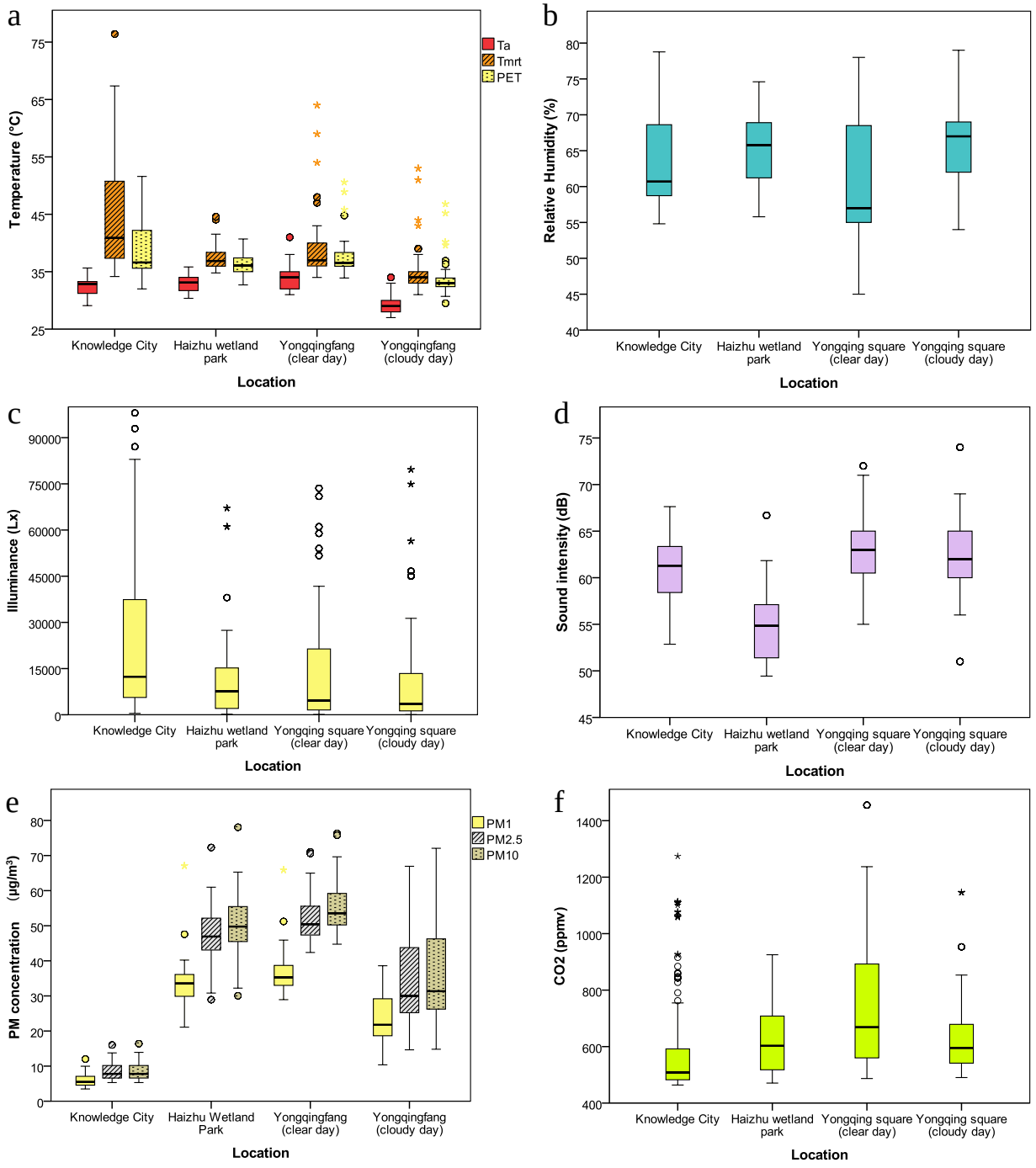


Fig. 5 Meteorological conditions during the survey campaign, showing a) air temperature (T_a), mean radiant temperature (T_{mrt}) and physiological equivalent temperature (PET), b) relative humidity, c) horizontal illuminance, d) sound intensity (dB), e) PM1, PM2.5, and PM10 concentration, f) CO2 concentration. The sunny and cloudy days for Yongqingfang were 29/9/2021 and 20/10/2021, respectively.

3.2 Effects of visual comfort on outdoor thermal comfort

A chi-square test of association indicated a significant association between perceived environmental quality (SSV, GSV, NAV, ADV, AQV), psychological indicators (irritability, tiredness) and thermal comfort indicators (TSV, TCV, TAV, and TTV) (n = 2,352) (Table 3). The exception was smelliness, which had no significant association with TCV, TAV and TTV (Table 3). After that, a Bonferroni-corrected z-test was used to determine which pairs of columns for a given row in the above-paired groups (thermal comfort indicators and perceived environmental quality indicators) are statistically different. Asterisks in Fig. 6 to Fig. 9 indicate the paired groups with significant differences.

There was a significant association between visual comfort and thermal comfort indicators, with a medium (ϕ_c = 0.38) to strong effect (ϕ_c =0.50). A change in the sun sensation vote (SSV) from very dark to very bright affects thermal comfort indicators. As SSV increased from 'very dark and dark' to 'bright and very bright', a significantly higher proportion of respondents felt hotter (Fig. 6a), more uncomfortable (Fig. 6b), more unacceptable (Fig. 6c) and more intolerable with the thermal environment (Fig. 6d). In particular, the percentage of 'very hot and extremely hot' vote (TSV= +4, +5), 'comfortable' vote (TCV=0), 'acceptable' vote (TAV=0) and 'tolerable' vote (TTV=0) was significantly different between SSV= 'slightly dark to slightly bright', SSV= 'bright and very bright' and the reference group (SSV= 'very dark or dark').

Similar to SSV, there was a strong association between glare sensation vote (GSV) and thermal comfort indicators. When the GSV changed from not glaring to very glaring, it led to a higher percentage of hotter thermal sensation (Fig. 6e), thermal discomfort (Fig. 6f), thermal unacceptability (Fig. 6g) and thermal intolerance vote (Fig. 6h). In brief, SSV and GSV had similar

effects on thermal comfort indicators and were generally strongly associated. Hot thermal sensation, thermal discomfort, thermal unacceptability and thermal intolerance were related to bright sun sensation and glaring sensation.

Table 3 Summary table of the chi-square test of association, showing the association between thermal comfort indicators (TSV, TCV, TAV, TTV) and various perceived outdoor environmental quality and psychological indicators (irritability and perceived tiredness) (n = 2,352).

		TSV		
Variable	χ^2	*df*	*p*	ϕ_c
SSV	702.8	6	<0.001	0.39
GSV	990.2	9	<0.001	0.38
NAV	52.1	9	<0.001	0.09
Smelliness	56.8	6	<0.001	0.11
ADV	42.3	9	<0.001	0.07
AQV	43.5	9	<0.001	0.08
Irritability	379.7	9	<0.001	0.23
Tiredness	106.1	9	<0.001	0.12
		TCV		
SSV	731.2	8	<0.001	0.39
GSV	1273.8	12	<0.001	0.43
NAV	173.1	12	<0.001	0.16
Smelliness	6.7	8	0.564	0.04
ADV	58.8	12	<0.001	0.09
AQV	60.4	12	<0.001	0.09
Irritability	686.8	12	<0.001	0.31
Tiredness	304.2	12	<0.001	0.21
		TAV		
SSV	467.3	2	<0.001	0.45
GSV	592.7	3	<0.001	0.50
NAV	34.8	3	<0.001	0.12
Smelliness	12.2	2	0.002	0.07
ADV	23.5	3	<0.001	0.10
AQV	31.4	3	<0.001	0.12
Irritability	235.9	3	<0.001	0.32
Tiredness	76.2	3	<0.001	0.18
		TTV		
SSV	727.7	8	<0.001	0.39
GSV	1374.4	12	<0.001	0.44
NAV	154.2	12	<0.001	0.15
Smelliness	9.3	8	0.319	0.04
ADV	40.4	12	<0.001	0.08
AQV	56.8	12	<0.001	0.09

Irritability	704.5	12	<0.001	0.32
Tiredness	337.4	12	<0.001	0.22

Note for abbreviation: TSV: thermal sensation vote, TCV: thermal comfort vote, TAV: thermal acceptance vote, TTV: thermal tolerance vote, SSV: sun sensation vote, GSV: glare sensation vote, NAV: noise annoyance vote, ADV: air dustiness vote, AQV: air quality vote.

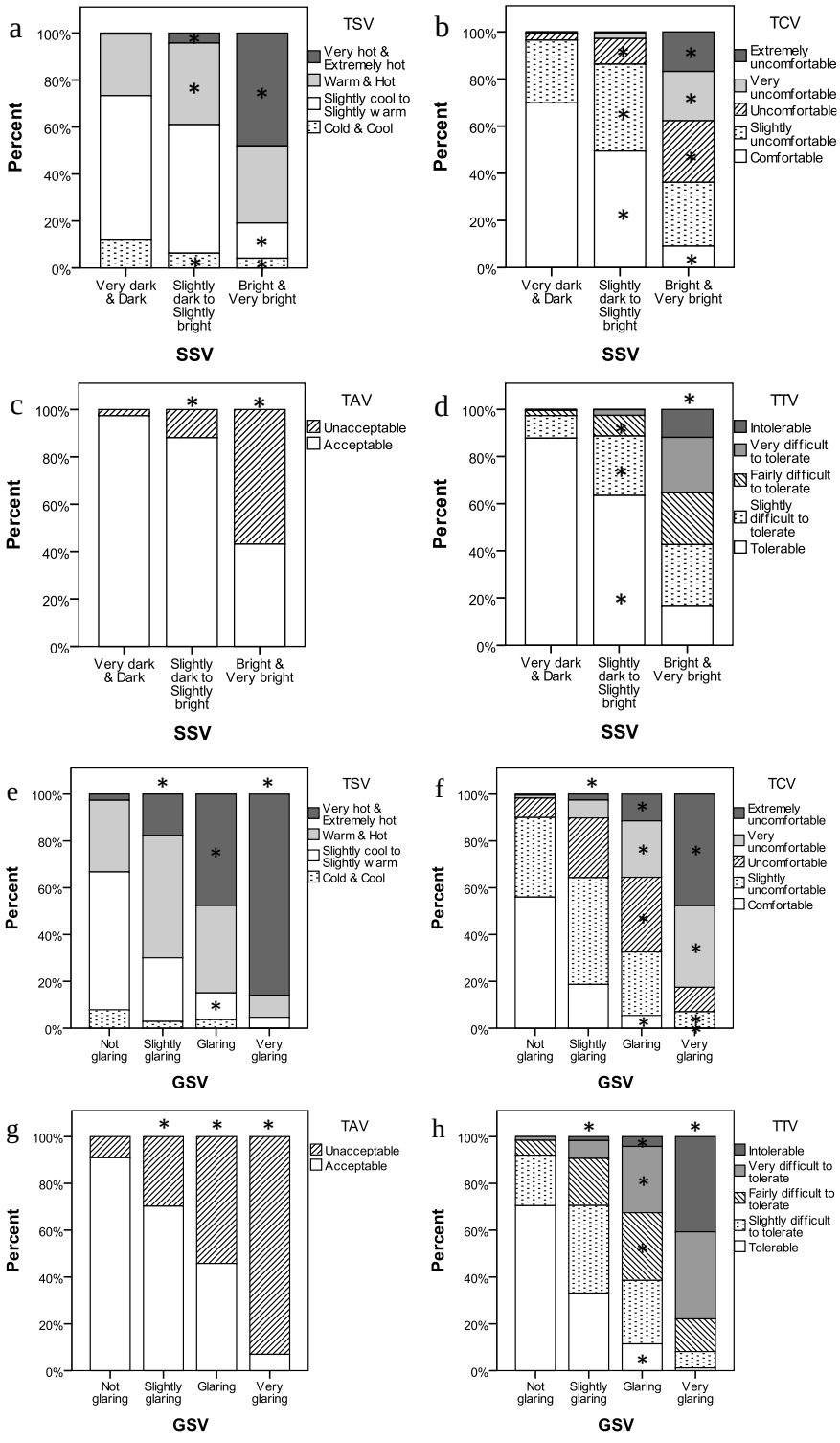


Fig. 6 The percentage of a) thermal sensation vote (TSV), b) thermal comfort vote (TCV), c) thermal acceptance vote (TAV) and d) thermal tolerance vote (TTV) for respondents with different sun sensation votes (SSV). The percentage of e) TSV, f) TCV, g) TAV and h) TTV for respondents

with different glare sensation votes (GSV). The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

The relationship between visual and thermal comfort is affected by the sun. To control for the confounding effect of the sun, we conducted a chi-square test of the association between visual comfort (SSV and GSV) and thermal comfort indicators (TSV and TCV) over similar horizontal illuminance ranges (see Table S2). In particular, we divided the survey samples into three horizontal illuminance groups based on Devaraj and Patel (2017). These three groups represent fully overcast skies to typical overcast skies at midday (1–2000 lx, n = 630), shade illuminated by a completely clear blue sky at midday (2000–20000 lx, n = 1106) and bright sunlight conditions (20000–110000 lx, n = 616). We found a significant association between SSV and TSV when the horizontal illuminance was 1-2000 lx (ϕ_c = 0.11, small effect), 2000-20000 lx (ϕ_c = 0.09, small effect) and 20000-110000 lx (ϕ_c = 0.43, large effect) (Fig. S1). There was also a significant association between SSV and TCV for 1-2000 lx (ϕ_c = 0.18, small effect), 2000-20000 lx (ϕ_c = 0.15, small effect) and 20000-110000 lx (ϕ_c = 0.43, large effect) (Fig. S2). For glare sensation, there was a significant association between GSV and TSV when the horizontal illuminance was 1–2000 lx (ϕ_c = 0.14, small effect), 2000–20000 lx (ϕ_c = 0.17, small effect) and 20000–110000 lx (ϕ_c = 0.41, large effect) (Fig. S3). Similarly, there was a significant association between GSV and TCV for 1-2000 lx (ϕ_c = 0.16, small effect), 2000-20000 lx (ϕ_c = 0.22, medium effect) and 20000-110000 lx (ϕ_c = 0.47, large effect) (Fig. S4). In brief, the relationship between visual and thermal comfort is apparent across different sunlight conditions, and the effect size is greater for bright sunlight conditions than overcast skies.

3.3 Effects of acoustic comfort on outdoor thermal comfort

A chi-square test of association indicated a significant association between noise annoyance vote (NAV) and thermal comfort indicators (Table 3), with a small effect (ϕ_c = 0.09-0.16). Compared with those who are not annoyed with the acoustic environment, a higher proportion of respondents felt hotter (TSV≥2) (Fig. 7a), uncomfortable (TCV>2) (Fig. 7b), unacceptable (TAV=1) (Fig. 7c) and intolerable (TTV>1) (Fig. 7d) when NAV was very annoying. However, the TSV and TAV proportion patterns were similar between NAV = not annoying and NAV = annoying, suggesting a lower noise annoyance level has a limited effect on TSV and TAV. Our results indicate that the proportion of thermal comfort indicators differs significantly with a higher NAV (i.e., very annoying). Moreover, NAV had a greater association with TCV and TTV than TSV and TAV.

We conducted a further chi-square test of association to investigate whether subjects with various NAV levels have different thermal comfort over similar PET ranges (see Table S2). The survey samples were divided into two PET groups based on Fang et al. (2019a): 29.1 °C–36.3 °C (warm sensation, n = 1358) and 36.3 °C–53.6 °C (hot sensation, n = 994). We found a significant association between NAV and TSV when PET was 36.3 °C–53.6 °C (ϕ_c = 0.10, small effect), but no significant association was found for a lower PET range (29.1°C–36.3°C) (p = 0.188) (Fig. S5). In comparison, there was a significant association between NAV and TCV when PET was 29.1°C–36.3°C (ϕ_c = 0.14, small effect) and 36.3 °C–53.6 °C (ϕ_c = 0.19, small effect) (Fig. S6). The association between noise annoyance and thermal sensation is more apparent over the higher PET range.

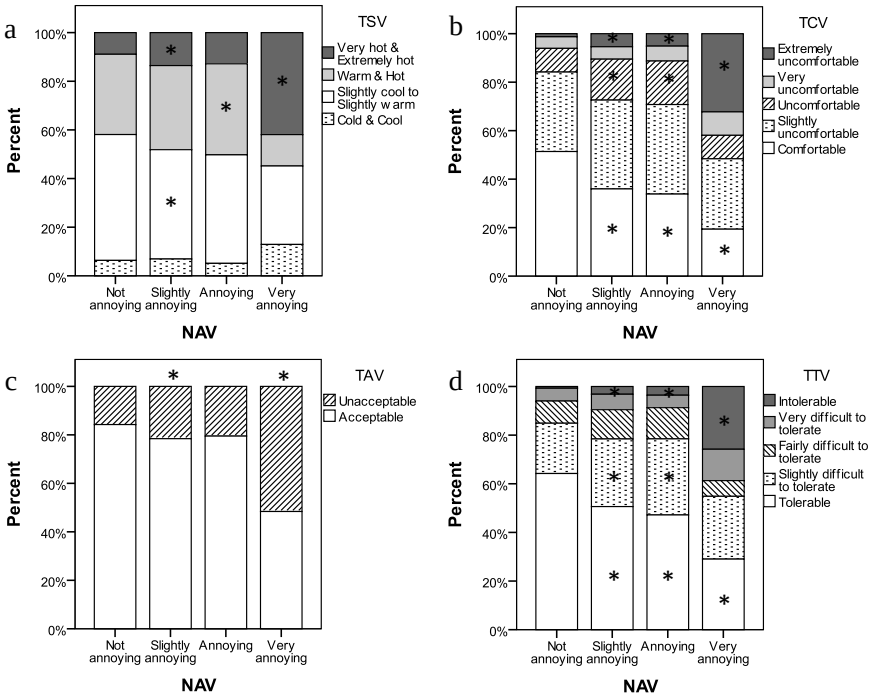


Fig. 7 The percentage of a) thermal sensation vote (TSV), b) thermal comfort vote (TCV), c) thermal acceptance vote (TAV) and d) thermal tolerance vote (TTV) for respondents with different noise annoyance votes (NAV). The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

3.4 Perceived air quality's effects on outdoor thermal comfort

A chi-square test of association indicated no significant association between smelliness and TCV, TAV and TTV, except for TSV ($\phi_c = 0.11$, weak effect) (Table 3). Compared with the 'not smelly' group, a smaller proportion of respondents felt warm and hot (TSV=2 and 3) when they perceived the environment as 'smelly and very smelly' (Fig. 8a). Overall, the olfactory comfort indicator (i.e., smelliness) has little association with other thermal comfort indicators.

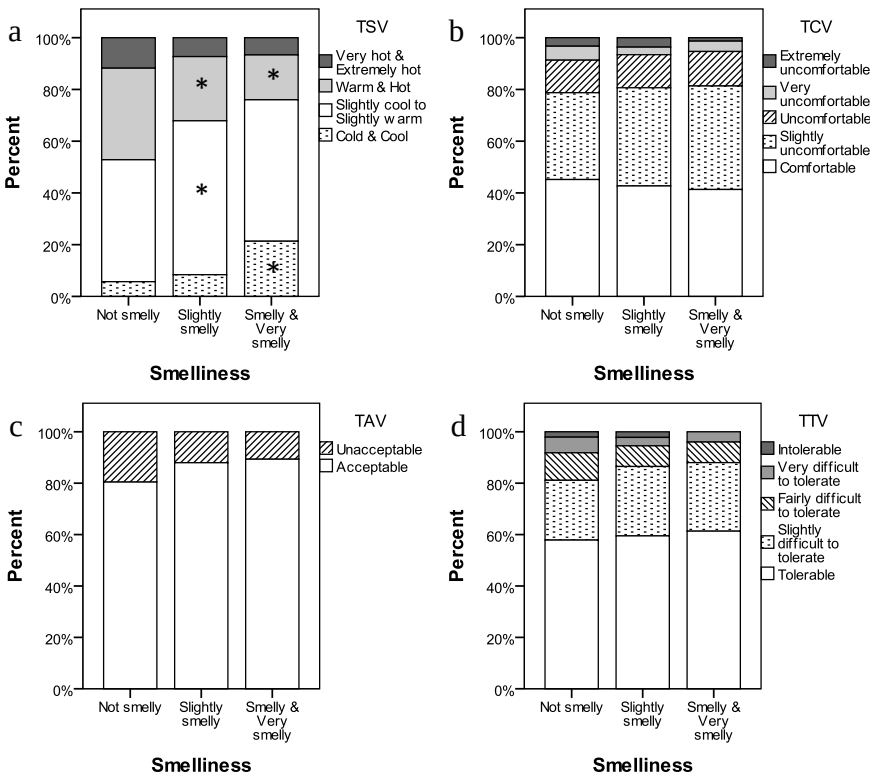


Fig. 8 The percentage of a) thermal sensation vote (TSV), b) thermal comfort vote (TCV), c) thermal acceptance vote (TAV) and d) thermal tolerance vote (TTV) for respondents with different levels of perceived smelliness. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

Unlike smelliness, a chi-square test of association showed a significant association between perceived air quality and thermal comfort ($\phi_c = 0.07–0.12$, weak effect) (Table 3). When ADV was very clean or AQV was very good, it was associated with a higher proportion of 'warm and hot' sensations (Fig. 9a and Fig. 9e), while associating with a higher proportion of respondents who found the thermal environment comfortable (Fig. 9b, Fig. 9f), acceptable (Fig. 9c, Fig. 9g) and tolerable (Fig. 9d, Fig. 9h). Our results suggest that very clean air condition is associated with a significant change in perceived thermal comfort. However, this is not the case for certain ADV (neither dusty nor clean and clean) and AQV (neither poor nor good and good).

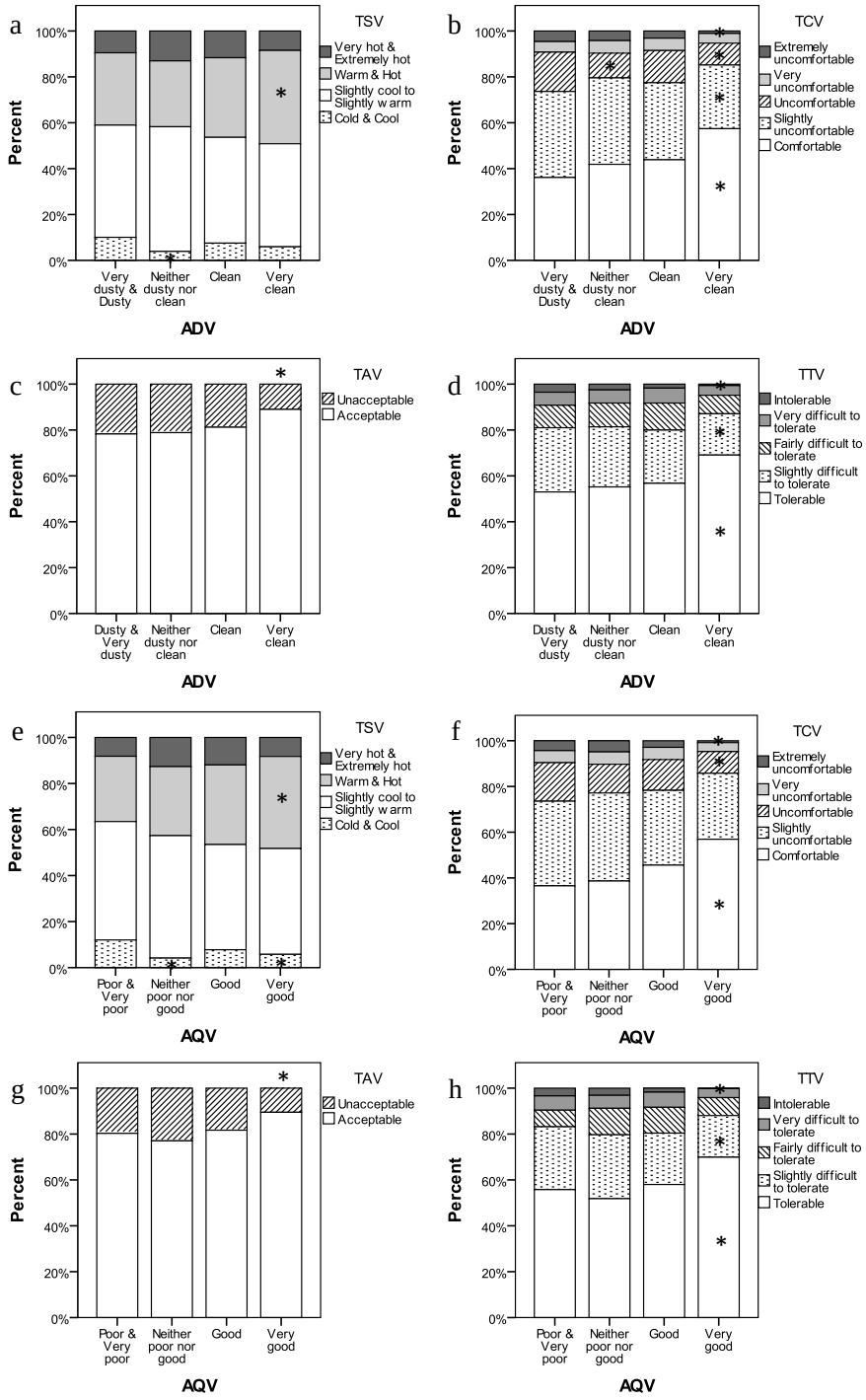


Fig. 9 The percentage of a) thermal sensation vote (TSV), b) thermal comfort vote (TCV), c) thermal acceptance vote (TAV) and d) thermal tolerance vote (TTV) for respondents with different air dustiness votes (ADV). The percentage of e) TSV, f) TCV, g) TAV and h) TTV for respondents with different air quality votes (AQV). The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

3.5 Effects of psychological status on outdoor thermal comfort

A chi-square test of associated indicated a significant association between respondents' psychological status (irritability and perceived tiredness) and thermal comfort indicators (Table 3). In particular, the effect size of irritability (ϕ_c = 0.23-0.32, small to medium effect) was greater than tiredness (ϕ_c = 0.12-0.22, small effect). The effect size of irritability and tiredness is greater for TCV, TAV and TTV (ϕ_c = 0.21-0.32) than TSV (ϕ_c = 0.12-0.23). When irritability changed from 'not irritated' to 'slightly irritated' and above, it led to a hotter thermal sensation (Fig. 10a), greater thermal discomfort (Fig. 10b), as well as more unacceptable (Fig. 10c) and intolerable with the thermal environment (Fig. 10d). Similar patterns are observed for tiredness (i.e., 'not tired' vs 'slightly tired' and above) (Fig. 10e-h). Based on the Bonferroni-corrected z-test results, respondents who were irritated and tired had significantly different proportions of thermal comfort indicators compared with those who were not irritated and not tired.

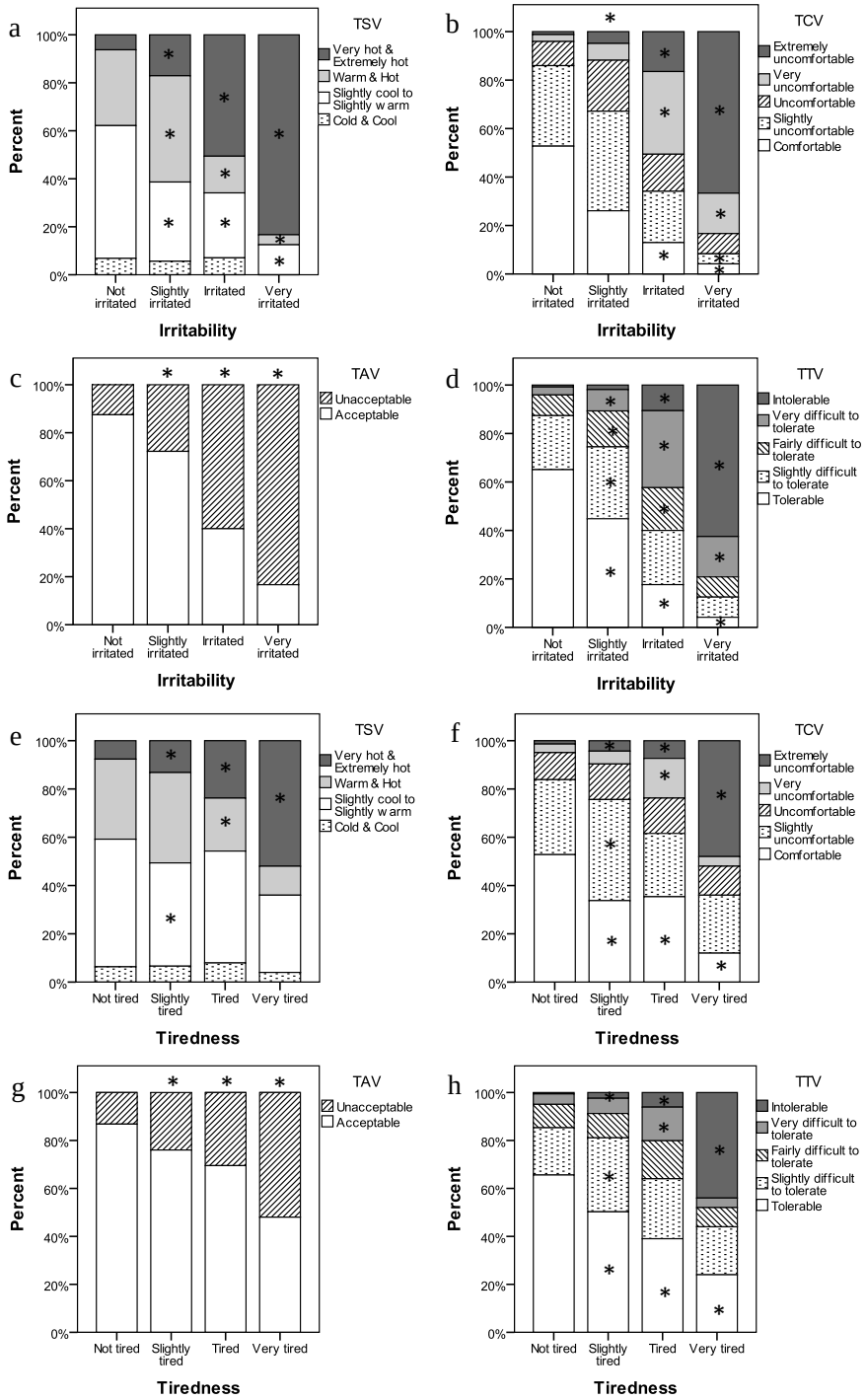


Fig. 10 The percentage of a) thermal sensation vote (TSV), b) thermal comfort vote (TCV), c) thermal acceptance vote (TAV) and d) thermal tolerance vote (TTV) for respondents with different irritability levels. The percentage of e) TSV, f) TCV, g) TAV and h) TTV for respondents with different perceived tiredness levels. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

3.6 Environmental and psychological indicators' impact on thermal comfort

We tested the multicollinearity and proportional odds assumption for our logistic regression model. All these models' tolerance values were above 0.1, indicating no multicollinearity. We found a significant Chi-square statistic ($p > 0.05$) in the test of parallel lines in SPSS (e.g., $p = 0.064$ for TCV and $p = 0.628$ for TTV). Therefore, our ordinal logistic regression (OLR) model also satisfies the proportional odds assumption. Since the TSV and TAV models violate the proportional odds assumption, we employed a binary logistic regression (BLR) model instead. As the independent variables are all categorical, the odds ratio (OR) value means how many times more likely the category is to result in respondents feeling hotter (or less comfortable, less acceptable, less tolerable) than the reference category. An OR value greater than 1 means that the respondent is more likely to feel hotter, uncomfortable, unacceptable, and intolerable with the thermal environment.

We used binary logistic regression (BLR) to assess the effects of the indicators of outdoor environmental quality and psychological status on TSV and TAV. The BLR model for TSV was statistically significant, $\chi^2(29, n = 2199) = 1149.6$, $p < 0.001$. The model explained 54.3% (Nagelkerke R^2) of the variance in TSV and correctly classified 80.1 % of cases. The BLR model for TAV was also statistically significant, $\chi^2(29, n = 2352) = 871.2$, $p < 0.001$. The model explained 50.3% (Nagelkerke R^2) of the variance in TAV and correctly classified 87.6 % of cases.

Ordinal logistic regression (OLR) was used to assess the effects of the same independent variables on TCV and TTV. Overall, the OLR model was statistically significant for TCV, $\chi^2 (29, n = 2352) = 1707.6$, $p < 0.001$. The model explained 56.2% (Nagelkerke R^2) of the variance in TCV and correctly classified 60.6 % of cases. The OLR model was also statistically significant for TTV, χ^2

(29, n = 2352) =1521.9, $p < 0.001$. The model explained 53.1% (Nagelkerke R^2) of the variance in TTV and correctly classified 65.5 % of cases.

Survey location, humidity sensation vote (HSV), wind sensation vote (WSV), shade conditions, SSV, GSV, irritability, and tiredness were significant predictors of all thermal comfort indicators (TSV, TCV, TAV and TTV) (Table A1 to Table A4). Gender was a significant predictor of TSV and TCV, but not TAV and TTV. In particular, females (OR_{TSV} = 9.69, $p < 0.001$, OR_{TCV} = 1.42, $p < 0.001$) felt hotter and more uncomfortable than males (Table A1 and Table A2). It means that females were 9.69 times more likely to feel hot than males. At the same time, females were 1.42 times more likely to feel uncomfortable than males.

Regarding shade conditions, those participants in exposed conditions were more likely to feel hotter (OR = 3.24, $p < 0.001$), more uncomfortable (OR = 1.65, $p < 0.001$), less acceptable (OR = 1.64, p = 0.009) and less tolerable (OR = 1.52, p = 0.001) than those in the shade. This result indicates the impact of shade on alleviating thermal discomfort.

For sun sensation (SSV), those who found the environment 'bright' (OR_{TSV} = 4.64, $p < 0.001$, OR_{TCV} = 6.20, $p < 0.001$) and 'very bright' (OR_{TSV} = 15.76, $p < 0.001$, OR_{TCV} = 34.24, $p < 0.001$) were more likely to feel hotter and more uncomfortable than those who found it 'dark and very dark' (Table A1 and Table A2). The 'bright' (OR_{TAV} = 8.06, $p < 0.001$, OR_{TTV} = 8.70, $p < 0.001$) and 'very bright' groups (OR_{TAV} = 42.32, $p < 0.001$, OR_{TTV} = 48.28, $p < 0.001$) were also more likely to feel unacceptable and intolerable than the 'dark and very dark' group (Table A3 and Table A4). These results indicate that the sun sensation vote significantly correlated with TSV, especially for the 'very bright' group.

For glare sensation (GSV), participants who found the environment 'slightly glaring' (OR_{TSV} = 3.88, $p < 0.001$, OR_{TCV} = 3.42, $p < 0.001$) and 'glaring and very glaring' (OR_{TSV} = 4.09, $p < 0.001$, OR_{TCV} = 6.05, $p < 0.001$) were more likely to feel hotter and more uncomfortable than those who found the environment 'not glaring' (Table A1 and Table A2). Furthermore, the 'slightly glaring' (OR_{TAV} = 3.39, $p < 0.001$, OR_{TTV} = 3.49, $p < 0.001$) and 'glaring and very glaring' groups (OR_{TAV} = 4.60, $p < 0.001$, OR_{TTV} = 6.81, $p < 0.001$) were more likely to find the environment unacceptable and intolerable than the 'not glaring' groups (Table A3 and Table A4). The OR value of the 'glaring and very glaring' group is higher than the 'slightly glaring', highlighting the effects of glare on thermal comfort. Like sun sensation, glare sensation is also a significant predictor of thermal comfort indicators.

The NAV made a unique statistically significant contribution to TSV, TCV and TTV, but not TAV. When participants found the acoustic environment 'slightly annoying' (OR_{TSV} = 1.34, p = 0.039, OR_{TCV} = 1.47, $p < 0.001$) or 'annoying and very annoying' (OR_{TSV} = 1.81, p = 0.009, OR_{TCV} = 1.61, p = 0.004) were likelier to feel hotter and more uncomfortable than those who are not annoyed (Table A1 and Table A2). The 'slightly annoying' (OR = 1.33, p = 0.009) and 'annoying and very annoying' groups (OR = 1.51, p = 0.017) were also more likely to feel intolerable than the 'not annoying' group (Table A4). Participants who were annoyed with the noise in the environment were more likely to feel hotter, more uncomfortable, and more intolerant, but noise annoyance was not associated with TAV.

The TSV, TAV and TTV were significantly associated with smelliness and AQV. However, the ADV is not significantly related to any thermal comfort indicators. When participants perceived the environment to be 'smelly', they felt cooler (OR = 0.65, p = 0.03), more acceptable (OR = 0.25,

$p < 0.001$), and more tolerable (OR = 0.61, $p < 0.001$) than the 'not smelly' group (Table A1, Table A3 and Table A4). The significant association between smelliness and TSV was also found in our chi-square test of association (Table 3). In contrast, participants who perceived the air quality to be 'good' (OR = 2.81, $p = 0.005$) and 'very good' (OR = 3.92, $p = 0.006$) were more likely to feel hotter than those who found the air quality 'bad and very bad' (Table A1). Moreover, those who found the air quality 'very good' were also more likely to feel more acceptable (OR = 0.27, $p = 0.035$) and more tolerable (OR = 0.29, $p = 0.001$) than the 'bad and very bad' group (Table A3 and Table A4). However, odour perception and perceived air quality indicators did not significantly predict TCV. These results suggest that good air quality perceptions were more likely to make participants feel hotter, but at the same time more acceptable and tolerable with the thermal environment.

Apart from outdoor environmental quality indicators, psychological status indicators (irritability and perceived tiredness) were also significantly related to thermal comfort indicators. When the participants were 'slightly irritated' ($OR_{TSV} = 2.32$, $p < 0.001$, $OR_{TCV} = 2.24$, $p < 0.001$) or 'irritated and very irritated' ($OR_{TSV} = 2.30$, $p = 0.042$, $OR_{TCV} = 5.88$, $p < 0.001$), they were more likely to feel hotter and more uncomfortable than those who were 'not irritated' (Table A1 and Table A2). Moreover, the 'slightly irritated' ($OR_{TAV} = 2.16$, $p < 0.001$, $OR_{TTV} = 1.56$, $p < 0.001$) and 'irritated and very irritated' participants ($OR_{TAV} = 8.09$, $p < 0.001$, $OR_{TTV} = 4.18$, $p < 0.001$) were likelier to find the environment unacceptable and intolerable (Table A3 and Table A4). As participants' irritability levels increased, the OR value of various thermal comfort indicators also increased.

Besides irritability, perceived tiredness is a significant predictor of thermal comfort indicators. When participants were ‘slightly tired’, they were more likely to feel hotter (OR = 1.54, $p = 0.003$), less comfortable (OR = 1.34, $p = 0.007$), unacceptable (OR = 1.61, $p = 0.006$) and intolerable (OR = 1.29, $p = 0.002$) than those who were ‘not tired’ (Table A1 to Table A4). When participants were ‘tired and very tired’, they were likelier to find the environment intolerable (OR = 1.81, $p = 0.002$) than the ‘not tired’ participants. However, the ‘tired and very tired’ group showed no significant correlation with TSV, TCV and TAV. Irritability and perceived tiredness are associated with different thermal comfort indicators, showing the impact of psychological status on adapting to the thermal environment.

In short, visual comfort indicators and psychological status indicators were significant predictors of all thermal comfort indicators. In contrast, acoustic comfort, perceived odour and air quality indicators only significantly predicted certain thermal comfort indicators. These results highlight the different dimensions of thermal comfort as an early warning signal for heat stress and climate adaptation.

4 Discussion

Our results indicate a cross-modal influence of visual, acoustic and air quality perception on thermal perception in the urban environment. We found that a higher sun sensation and glare sensation affect people's thermal perception. A more annoying noise perception can lead to higher thermal discomfort. While perceived odour has limited influence on thermal perception, a better air quality perception leads to better thermal comfort. Psychological status (e.g., higher irritability and perceived tiredness) is also associated with higher levels of thermal discomfort. Females are more likely to feel hotter and more uncomfortable than males because females have a higher sensitivity to changes in environmental conditions (Chen et al., 2021) and are less tolerant to heat (Tung et al., 2014). Our findings highlight the need to consider different indicators of outdoor environmental quality and psychological status to assess individual climate change adaptation to increasing urban heat.

4.1 Association between visual comfort and thermal comfort

We found that higher sun sensation led to worse thermal comfort perception (Fig. 6a-d, Fig. S1-S2). Similarly, a Chinese study shows that an increase in visual perceptions increases thermal sensation (Liu et al., 2023a). However, past studies from Italy (Cureau et al., 2022) and Indonesia (Nitidara et al., 2022) indicate that people feel more comfortable when they perceive the environment to be brighter. The discrepancy between past findings and our results could be related to the differences in the study site, in which people might prefer brighter visual sensation in parks compared with urban squares. The relationship between visual and thermal comfort also depends on the level of heat stress (Geng et al., 2022; Lam et al., 2020). For example, the thermal sensation vote is positively correlated with the sun sensation vote, and thermal perception is significantly

associated with sunlight preference (Lam et al., 2020). The sunlight exposure produces thermal and visual sensations at the same time. Sunlight, which is associated with warmth and heat, could activate the human's hot emotional system (Xu & Labroo, 2014). The heat of sunlight is also received by the skin (Nitidara et al., 2022). We found that the effect size of the relationship between sun sensation vote and thermal sensation vote is greater for bright sunlight conditions than overcast skies (Fig. S1). This finding is supported by a study in Beijing, China (Liu et al., 2023b), which indicate that high solar radiation is associated with a high level of illuminance and causes people to experience strong local heating. Past studies suggest that people could feel hotter at a higher solar radiation intensity at a similar thermal index range (Lam & Hang, 2017). Light also affects human melatonin levels. Melatonin could affect thermoregulation (Saarela & Reiter, 1994) by changing the heat loss induction through skin temperature (Egan et al., 2005) and core body temperature (Cajochen et al., 2005), which affects thermal sensation (Wang et al., 2007). In brief, both visual and non-visual pathways of light could affect thermal comfort.

Apart from sun sensation, glare perception also affects people's thermal comfort perception (Fig. 6e-h, Fig. S3-S4). This result is consistent with a study in central Italy which indicates that glare effects could affect pedestrian comfort (Rosso et al., 2016). Glare refers to the excessive brightness level within the field of vision (Quek et al., 2021). The source of glare in the urban environment could be a reflection of building façade materials (Achsani & Wonorahardjo, 2020). Excessive direct sunlight in outdoor environments would need to be addressed to enhance outdoor thermal and visual comfort (Pan & Du, 2021). Therefore, our findings emphasize that building design would need to avoid glare in street canyons to improve pedestrian comfort level.

4.2 Association between acoustic comfort and thermal comfort

When people are more annoyed with the noise in the environment, it can make people feel more irritated and hotter than those who are not annoyed with the acoustic environment (Fig. 7). Our findings are in good agreement with a Hong Kong study by Lau and Choi (2021), who state a perceptually quiet environment could lead to a higher thermal tolerance. Similarly, the level of thermal comfort reduces with a high noise level (Guan et al., 2020a; Yin et al., 2022b) and greater dissatisfaction with the acoustic quality (Lau & Choi, 2021). Increasing sound pressure levels can also affect the sense of irritation rating and subsequently thermal discomfort (Chen et al., 2023; Guan et al., 2020b).

We found that the association between NAV and TSV was only significant over higher PET range (Fig. S5). This finding is supported by a Chinese study by Guan et al. (2020a), who demonstrate a more noticeable impact of noise on thermal comfort in a hotter environment (e.g., > 30 °C). In warm conditions, a high noise level increases the heart rate, which increases thermal discomfort (Guan et al., 2020a). Yet, the skin temperature and fingertip blood perfusion do not differ under different sound environments (Guan et al., 2020b). These findings suggest that thermos-physiology affects the relationship between the acoustic environment and thermal comfort in various ways and warrants further study.

4.3 Association between perceived air quality and thermal comfort

We found that perceived odour has little association with TCV, TAV and TTV (Fig. 8). However, this result contrasts with past Chinese studies that demonstrate a possible effect of perceived odour on thermal comfort (Jia et al., 2023; Liu et al., 2023a). Odour can be associated with heat or cold

properties relating to traditional conscious knowledge, which affects people's heat sensation (Wnuk et al., 2017). Moreover, odour can activate the trigeminal receptors in the human nose, resulting in the sensation of temperature changes (Gingras-Lessard & Frasnelli, 2016). In particular, Chang et al. (2023) suggest that increasing the fragrance comfort vote can alleviate thermal discomfort during summer. The discrepancy between our findings and past studies might be due to the differences in odour sources, such as green spaces on a university campus in Chang et al. (2023) and outdoor urban environment in this study. The questions on olfactory comfort also differ between Chang et al. (2023) (fragrance comfort vote) and our study (smelliness vote), which could explain the differences in results.

A better air quality vote and air dustiness vote are more likely to be associated with better thermal comfort and tolerance levels in this study (Fig. 9). Similar findings have been reported in a Hong Kong study by Ma et al. (2021) and a Dutch study by Peng et al. (2019), which indicate links between perceived air quality and pedestrian comfort. An increased PM2.5 concentration increased thermal sensation and reduced thermal comfort (Yin et al., 2022a). The amount of pollutant concentration would be related to the wind velocity in the urban street canyon, which also affects thermal comfort indices (Li et al., 2022). Perceived air quality affects environmental perception (Peng et al., 2021b) and the acceptability and need satisfaction of outdoor activity, subsequently affecting pedestrians' outdoor comfort (Peng et al., 2021a). In short, perceived outdoor air quality would need to be considered when assessing outdoor thermal comfort.

4.4 Association between psychological status and outdoor thermal comfort

Higher irritability level is more likely to be associated with hotter sensations and worse thermal comfort levels (Fig. 10a-d). This finding agrees with a Dutch study by Peng et al. (2021a), who

demonstrated that emotional state could affect thermal comfort in urban public space. Similarly, psychological cognition, like the sense of pleasure, communication, and satisfaction, positively affects thermal comfort (Liu et al., 2023a). Positive emotion can also decrease as thermal sensation increases (Niu et al., 2023). Increasing negative emotion can increase thermal sensation vote, which varies with different ranges of PET (Zhang et al., 2021). Furthermore, higher ambient temperature can increase negative emotions and tiredness (Noelke et al., 2016). A Chinese indoor study has also found that people tend to feel warmer in a boring state compared with a neutral and joyful state when they are sitting or standing, as blood pressure and heart rate increase during a boring state (Wang & Liu, 2020). However, they found that the participants' emotional state had no significant effect on thermal comfort during exercise. Although our results differ slightly from those of Wang and Liu (2020), the difference is possibly due to less extreme positive or negative emotions experienced in the indoor study.

A higher perceived tiredness level is associated with worse thermal comfort perceptions (Fig. 10e-h). This finding confirms another study in Guangzhou, China, which shows perceived tiredness as a predictor of outdoor thermal comfort during exercise (Lam et al., 2021b). Perceived tiredness is related to the perception of effort in exercise tolerance in the psychobiological model (Smirmaul et al., 2013). When people feel tired, it could reduce their ability to cope with heat stress, thus causing them to feel hotter and more uncomfortable in a hot environment. In short, psychological adaptation (such as emotional status and perceived tiredness) plays a major role in how people cope with heat stress.

4.5 Implications for urban planning and public health policies

Most of the current urban planning practice focuses on one aspect of the outdoor environmental quality. However, our results suggest that a multi-domain analysis considering thermal, visual, acoustic and air quality could provide a deeper insight into the impact of urban design on human comfort. What stands out in this work is that multiple environmental quality variables would affect pedestrian comfort, which leads to some practical advice on incorporating multi-domain comfort assessment in sustainable urban planning.

First, urban planners would need to consider the effects of urban vegetation on different outdoor environmental quality factors. Our findings indicate that the shortcomings in one environmental factor (e.g., thermal) can be offset by improving other factors (e.g., visual), which is in agreement with Nitidara et al. (2022). Urban vegetation not only mitigates urban heat (Kondo et al., 2021; Shooshtarian et al., 2018) and promotes walkability (Xiong & He, 2024), but it also increases a location's visual and aesthetic appeal and lowers people's thermal sensation (Lau & Choi, 2021). Moreover, pedestrian comfort levels are influenced by perceived air quality (Ma et al., 2021; Peng et al., 2019), which can be related to the amount of urban vegetation. The air quality in parks is perceived to be better than the surrounding built environment (Rosso et al., 2022), which is supported by a higher particulate matter concentration in roads than urban green space (Wang et al., 2022). The dust retention effects of plants and the negative air ions released by vegetation can remove particulate matter and improve air freshness (Jiang et al., 2021). However, excessive vegetation in high-density urban environments could reduce ventilation and increase the concentration of air pollutants (Fang et al., 2021). The effect of vegetation on air quality depends on the species and air pollution level (Janhäll, 2015). Furthermore, street trees could affect both

the thermal environment and air quality depending on the planting density and leaf area index (Li et al., 2023). Briefly, our findings suggest that urban greenery management needs to consider the trade-off between various outdoor environmental quality factors, including thermal comfort, visual comfort and air quality.

Second, urban planners should promote natural sound types in cities to improve acoustic comfort, which subsequently affects outdoor thermal comfort (Du et al., 2023). However, the effects of acoustic perception on thermal comfort differ for various sound types (Chen et al., 2022a). Acoustic comfort can be related to sound pressure level and spectrum (manifested in different sound types) (Cureau et al., 2022; Du et al., 2023). For example, traffic noise can lower thermal comfort in summer (Jin et al., 2020a; Nitidara et al., 2022). In contrast, certain natural sound types (e.g., birdsong, water, background music) are perceived to be favourable by humans and even positively influence thermal comfort (Chen et al., 2022a; Geng et al., 2022; Guan et al., 2020b). Water sounds from fountains can also mask the road traffic noise (Axelsson et al., 2014) and increase people's comfort level. As birdsong and music have more middle- and high-frequency components and a higher variety variability in sound, people appear to have a higher acceptance of these sounds (Jin et al., 2020b). Moreover, positive sound sources improve the overall ratings on comfort, irritation and satisfaction (Chen et al., 2023), as well as emotional state (Ito et al., 2023). Depending on the sound type, people can be more tolerant of a higher sound pressure if they find it pleasant. As such, urban planner should introduce more natural sound in the urban landscape to improve acoustic comfort, which can reduce thermal discomfort.

Third, urban planners would need to recognise the influence of the audio-visual context of a location on the emotional state of outdoor users. In particular, the congruency between the audio

and visual components of a location would enhance the pleasantness of natural sounds (Hong et al., 2020). Interaction with nature-based solutions also benefits mental health (Sharifi et al., 2021). Urban planners should promote natural elements in the urban landscape, such as natural sound and openness, to foster psychological restoration and improve mental well-being (Jeon et al., 2023; Yin et al., 2023). Images and sound in the natural environment can also induce physiological comfort and subjective restorativeness (Li et al., 2021). Furthermore, the audio-visual interaction relating to urban vegetation can positively affect people's emotional state, reducing anxiety and tiredness (Elsadek et al., 2019). Moreover, more positive emotion (e.g., cheerful, relaxed) is reported in the waterscape than in the urban built environment (Yan et al., 2023), which is associated with more negative emotion (e.g., anxiety, tiredness) (Zhang et al., 2019). The lower anxiety and perceived stress level in peri-urban forests is associated with a lower thermal sensation vote (Gillerot et al., 2024), further supporting our findings. In brief, urban planners should consider multisensory interaction, environmental perception and users' psychology status in managing street greenery and waterbody, which enhance the overall pedestrian comfort.

Apart from informing urban planning practices, our findings have implications for designing personalized early warning systems for vulnerable urban populations. There has been a shift of focus from a population-level heat-health warning system to a personalized heat-health exposure assessment (Nazarian & Lee, 2021) and warning system (Kuras et al., 2017; Morris et al., 2021). Past heat-health studies highlight the heterogeneous characteristics of the urban population (Forceville et al., 2024; Harlan et al., 2013; Tuholske et al., 2021). Moreover, people's age, education level and income affect their heat adaptation (He et al., 2022). Hence, it is necessary to provide targeted warning information to suit individual needs (Jay et al., 2021). The World Health Organization sees thermal comfort as a link to protect vulnerable groups from extreme

temperatures (Ormandy & Ezratty, 2012). Thermal comfort can be seen as an early warning signal to prevent the development of more serious heat-related illnesses (Lam et al., 2021a). This study underscores the importance of considering the multisensory influence on thermal comfort, thus providing a holistic evaluation of people's adaptation to heat stress and informing adaptation measures to produce health co-benefits (Sharifi et al., 2021). Smartphone apps like the ClimApp include weather variables and user input to predict individual thermal stress (Eggeling et al., 2022; Folkerts et al., 2021). Our study emphasises the need to incorporate other environmental variables (e.g., illuminance, noise and air quality) (Matzarakis, 2020) in such smartphone apps. Developing the smartphone app and personalized warning system would require input from citizen-based monitoring projects on different environmental variables, such as air temperature (Lam et al., 2021c), light intensity (Cabrera et al., 2021), noise (Longo et al., 2020), and air quality (Mahajan et al., 2022). Feedback from end-users would also improve the app design and functions (Neset et al., 2021). The multisensory interaction in the outdoor environment is key to improving people's thermal adaptation in light of climate change. Thus, the present findings indicate that a comprehensive early warning system requires not only heat exposure but also exposure to other environmental quality variables, which would increase the resilience of the urban population to urban heat.

4.6 Strengths and limitations of this study

The study contributes to the field in two ways. First, this study has highlighted the importance of the cross-modal effect of visual comfort, acoustic comfort and perceived air quality on outdoor thermal comfort. Our method of thermal walk and multi-domain comfort surveys could be applied to similar studies in other climate zones. The value of our work lies in informing appropriate urban

design interventions to reduce undesirable exposure to environmental variables. The shortcomings in one environmental factor (e.g., noise) can be counteracted by improving other factors (e.g., visual). Therefore, climate-resilient urban design that considers multisensory interaction is critical to improving the resilience of urban populations to future climate change. Second, our findings emphasise the need for a paradigm shift from considering only heat stress to incorporating multisensory comfort and psychological status in early warning systems. These heat-health warning systems would need to consider other forms of environmental exposure, such as undesirable levels of light and glare, noise and air pollution. Our data suggest that self-report thermal discomfort (considering multisensory interaction and psychological status) can be an early warning signal to prevent the development of more serious heat-related illnesses. In particular, these personalised heat exposure data provide higher resolution information for developing an effective early warning system tailored towards different individuals.

This study has several limitations. First, the participants of this study are university students, who are mainly young and healthy adults. The environmental perception of other vulnerable groups, such as the elderly, would likely differ (Baquero Larriva & Higueras, 2020). Therefore, it would be worthwhile to include other populations in future studies to compare the impact of age groups on multisensory interaction and comfort perception. Second, due to the limitation in recruiting participants, we had one participant from the Gansu Province. Future studies could ensure that all participants come from the same geographical region. Third, this study mainly concerns the subjective perception of environmental quality in outdoor environments. We acknowledge that physiological parameters could affect people's comfort during exercise (Lam et al., 2021b). As this study focuses on multisensory interaction, it is beyond the scope of this study to examine the impact of physiological variables on outdoor thermal comfort. Fourth, we mainly consider

irritability and perceived tiredness to represent participants' emotional status. We recognise that other psychological adaptations and emotions can affect people's comfort perception and are worth investigating further.

5 Conclusions

This study examined the effects of visual comfort, acoustic comfort, perceived air quality and psychological status on outdoor thermal comfort. We used mobile transect measurements and thermal walks in three study sites in Guangzhou, China, as a case study. The following conclusions are drawn.

1) Sun sensation vote and glare sensation vote were significant predictors of thermal comfort indicators. The 'very bright' (OR = 15.76–48.28) and 'glaring and very glaring' groups (OR = 4.09–6.81) were more likely to feel hotter, more uncomfortable, unacceptable and intolerable than the 'dark' and 'not glaring' groups.
2) The noise annoyance vote was a significant predictor of thermal comfort indicators, except for the thermal acceptance vote. People who found the noise level 'annoying and very annoying' (OR = 1.51–1.81) were likelier to feel hotter, more uncomfortable and intolerable than those who were not annoyed.
3) The air quality vote was a significant predictor of outdoor thermal comfort. People who perceived the air quality as 'very good' were likelier to feel hotter (OR = 3.92), acceptable (OR = 0.27) and tolerable (OR = 0.29) than the 'bad and very bad' group. However, odour perception (smelliness) and perceived air quality indicators did not significantly predict TCV.

4) Our results demonstrate the impact of psychological status on adapting to the thermal environment. The ‘irritated and very irritated’ (OR = 2.30–8.09) and ‘slightly tired’ groups (OR = 1.29–1.61) were more likely to feel hotter, uncomfortable, unacceptable and intolerable than the ‘not irritated’ and ‘not tired’ groups.

In terms of future research directions, studies can conduct physiological measurements like heart rate and skin temperature (Nazarian et al., 2021) and compare the physiological data with subjective comfort perception under different urban environments. Moreover, future studies can incorporate other aspects like electroencephalogram (Ma et al., 2023) on emotional and brainwave responses. It is also worthwhile to analyse the impact of emotion on comfort perception (Gillerot et al., 2024) using the Profile of Mood States (POMS) (McNair et al., 1992) or Positive and Negative Affective Schedule (PANAS) (Watson et al., 1988), which provides more comprehensive information on people’s emotional status. Overall, the significance of this study lies in informing climate-resilient urban design and heat-health warning systems that incorporate multisensory interaction and people’s psychological status.

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University – project number 2018 – No. 041. The data obtained by the survey in this study were anonymized. Informed consent was obtained from all individual participants in this study. We would also like to thank Dr Hong Ling for providing the portable air quality sensors for our monitoring campaign.

7 Appendix A

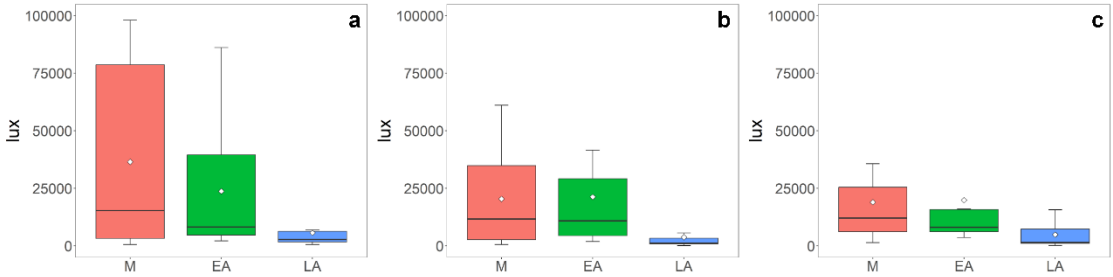


Fig. A1 Horizontal illuminance for each period on the three walk experiment routes, showing a) Knowledge City, b) Yongqingfang, and c) Haizhu Wetland. M, EA, and LA refer to three periods from 09:30-11:00. 13:30-15:00, 16:30-18:00, respectively.

Table A1 Multivariate binary logistic results of thermal sensation vote (TSV), perceived environmental quality and psychological status. The TSV is divided into two groups (slightly cool to slightly warm: -1 to +1, and warm to extremely hot: +2 to +5). The table shows the unstandardized coefficients (*B*), standard error (*SE*), wald statistics (Wald), p-value (*p*), odds ratio (OR), and 95% confidence intervals (95% CI). An OR value greater than 1 means that people are more likely to feel warm to extremely hot.

Variable	*B*	*SE*	Wald	*p*	OR	95% CI
Location						
Knowledge City	0[a]				1	
Haizhu Wetland Park	0.53	0.21	6.48	0.011*	1.69	(1.13, 2.54)
Sunny Yongqingfang	1.05	0.18	36.00	<0.001*	2.86	(2.03, 4.04)
Cloudy Yongqingfang	-0.36	0.18	3.90	0.048*	0.70	(0.49, 1.00)
Gender						
Male	0[a]				1	
Female	2.27	0.15	237.11	<0.001*	9.69	(7.26, 12.94)
Humidity Sensation Vote (HSV)						
Dry & Very dry	0[a]				1	
Neutral	-0.64	0.17	13.84	<0.001*	0.53	(0.38, 0.74)
Humid & Very humid	-0.44	0.20	5.00	0.025*	0.65	(0.44, 0.95)
Wind Sensation Vote (WSV)						
No wind	0[a]				1	
Light wind	-1.34	0.13	103.35	<0.001*	0.26	(0.20, 0.34)
Windy	-1.67	0.26	40.38	<0.001*	0.19	(0.11, 0.31)
Shade conditions						
Shaded	0[a]				1	
Exposed	1.18	0.16	55.73	<0.001*	3.24	(2.38, 4.41)
Sun Sensation Vote (SSV)						
Dark & Very dark	0[a]				1	
Slightly dark	0.43	0.23	3.44	0.064	1.53	(0.98, 2.41)
Neutral	0.56	0.21	6.95	0.008*	1.74	(1.15, 2.64)

Slightly bright	0.98	0.26	13.80	<0.001*	2.67	(1.59, 4.48)
Bright	1.53	0.34	20.25	<0.001*	4.64	(2.38, 9.06)
Very bright	2.76	0.59	22.19	<0.001*	15.76	(5.00, 49.63)
Glare Sensation Vote (GSV)						
Not glaring	0[a]				1	
Slightly glaring	1.36	0.20	44.33	<0.001*	3.88	(2.60, 5.78)
Glaring & Very glaring	1.41	0.38	13.42	<0.001*	4.09	(1.93, 8.70)
Noise Annoyance Vote (NAV)						
Not annoying	0[a]				1	
Slightly annoying	0.29	0.14	4.28	0.039*	1.34	(1.02, 1.77)
Annoying & Very annoying	0.59	0.23	6.82	0.009*	1.81	(1.16, 2.82)
Smelliness						
Not smelly	0[a]				1	
Smelly	-0.42	0.20	4.71	0.030*	0.65	(0.45, 0.96)
Air Dustiness Vote (ADV)						
Dusty & Very dusty	0[a]				1	
Neither dusty nor clean	0.16	0.23	0.48	0.490	1.17	(0.75, 1.83)
Clean	0.26	0.30	0.76	0.384	1.30	(0.72, 2.35)
Very clean	0.46	0.44	1.09	0.297	1.58	(0.67, 3.76)
Air Quality Vote (AQV)						
Bad & Very bad	0[a]				1	
Neither poor nor good	0.35	0.29	1.52	0.218	1.42	(0.81, 2.50)
Good	1.03	0.36	8.05	0.005*	2.81	(1.38, 5.74)
Very good	1.37	0.49	7.67	0.006*	3.92	(1.49, 10.29)
Irritability						
Not irritated	0[a]				1	
Slightly irritated	0.84	0.17	25.62	<0.001*	2.32	(1.68, 3.22)
Irritated & Very irritated	0.83	0.41	4.15	0.042*	2.30	(1.03, 5.12)
Tiredness						
Not tired	0[a]				1	
Slightly tired	0.43	0.14	8.93	0.003*	1.54	(1.16, 2.04)
Tired & Very tired	-0.05	0.26	0.04	0.849	0.95	(0.57, 1.59)

Note [a] = reference category, * $p < 0.05$, CI = confidence interval for odds ratio (OR)

Table A2 Multivariate ordinal logistic results of thermal comfort vote (TCV), perceived environmental quality and psychological status. The TCV is divided into five groups (comfortable: 0, slightly uncomfortable: +1, uncomfortable: +2, very uncomfortable: +3, and extremely uncomfortable: +4). The table shows the unstandardized coefficients (*B*), standard error (*SE*), wald statistics (Wald), p-value (*p*), odds ratio (OR), and 95% confidence intervals (95% CI). An OR value greater than 1 means that people are more likely to feel more uncomfortable.

Variable	*B*	*SE*	Wald	*p*	OR	95% CI
Location						
Knowledge City	0[a]				1	
Haizhu Wetland Park	1.04	0.15	46.00	<0.001*	2.84	(2.10, 3.84)
Sunny Yongqingfang	0.58	0.13	21.57	<0.001*	1.79	(1.40, 2.29)
Cloudy Yongqingfang	-0.44	0.13	10.56	0.001*	0.65	(0.50, 0.84)
Gender						
Male	0[a]				1	
Female	0.35	0.10	13.19	<0.001*	1.42	(1.18, 1.72)
Humidity Sensation Vote (HSV)						
Dry & Very dry	0[a]				1	
Neutral	-0.48	0.12	16.26	<0.001*	0.62	(0.49, 0.78)
Humid & Very humid	-0.40	0.14	8.35	0.004*	0.67	(0.51, 0.88)
Wind Sensation Vote (WSV)						
No wind	0[a]				1	
Light wind	-1.45	0.10	209.52	<0.001*	0.23	(0.19, 0.29)
Windy	-2.61	0.21	152.12	<0.001*	0.07	(0.05, 0.11)
Shade conditions						
Shaded	0[a]				1	
Exposed	0.50	0.12	17.08	<0.001*	1.65	(1.30, 2.09)
Sun Sensation Vote (SSV)						
Dark & Very dark	0[a]				1	
Slightly dark	0.67	0.19	12.16	<0.001*	1.96	(1.34, 2.87)
Neutral	0.86	0.18	23.71	<0.001*	2.37	(1.67, 3.35)
Slightly bright	0.98	0.21	21.10	<0.001*	2.66	(1.75, 4.05)
Bright	1.82	0.25	52.42	<0.001*	6.20	(3.78, 10.16)
Very bright	3.53	0.33	114.30	<0.001*	34.24	(17.91, 65.44)
Glare Sensation Vote (GSV)						
Not glaring	0[a]				1	
Slightly glaring	1.23	0.15	66.19	<0.001*	3.42	(2.54, 4.60)
Glaring & Very glaring	1.80	0.23	58.74	<0.001*	6.05	(3.82, 9.58)
Noise Annoyance Vote (NAV)						
Not annoying	0[a]				1	
Slightly annoying	0.38	0.10	13.84	<0.001*	1.47	(1.20, 1.80)
Annoying & Very annoying	0.48	0.16	8.46	0.004*	1.61	(1.17, 2.23)
Smelliness						
Not smelly	0[a]				1	
Smelly	-0.25	0.14	3.37	0.066	0.78	(0.59, 1.02)
Air Dustiness Vote (ADV)						

Dusty & Very dusty	0[a]				1	
Neither dusty nor clean	0.10	0.16	0.35	0.557	1.10	(0.80, 1.51)
Clean	0.39	0.22	3.32	0.069	1.48	(0.97, 2.26)
Very clean	-0.40	0.32	1.59	0.207	0.67	(0.36, 1.25)
Air Quality Vote (AQV)						
Bad & Very bad	0[a]				1	
Neither poor nor good	0.05	0.20	0.07	0.794	1.05	(0.71, 1.56)
Good	-0.39	0.25	2.39	0.122	0.68	(0.41, 1.11)
Very good	-0.58	0.35	2.76	0.096	0.56	(0.28, 1.11)
Irritability						
Not irritated	0[a]				1	
Slightly irritated	0.81	0.12	43.85	<0.001*	2.24	(1.77, 2.85)
Irritated & Very irritated	1.77	0.25	51.67	<0.001*	5.88	(3.63, 9.53)
Tiredness						
Not tired	0[a]				1	
Slightly tired	0.29	0.11	7.22	0.007*	1.34	(1.08, 1.65)
Tired & Very tired	0.32	0.19	2.89	0.089	1.37	(0.95, 1.98)

Note [a] = reference category, * $p < 0.05$, CI = confidence interval for odds ratio (OR)

Table A3 Multivariate binary logistic results of thermal acceptance vote (TAV), perceived environmental quality and psychological status. The TAV is divided into two groups (acceptable: 0, and unacceptable: +1). The table shows the unstandardized coefficients (*B*), standard error (*SE*), wald statistics (Wald), p-value (*p*), odds ratio (OR), and 95% confidence intervals (95% CI). An OR value greater than 1 means that people are more likely to feel unacceptable.

Variable	*B*	*SE*	Wald	*p*	OR	95% CI
Location						
Knowledge City	0[a]				1	
Haizhu Wetland Park	1.17	0.25	21.38	<0.001*	3.24	(1.97, 5.33)
Sunny Yongqingfang	0.75	0.21	12.51	<0.001*	2.11	(1.39, 3.19)
Cloudy Yongqingfang	-0.28	0.25	1.27	0.259	0.76	(0.47, 1.23)
Gender						
Male	0[a]				1	
Female	-0.02	0.16	0.02	0.902	0.98	(0.72, 1.34)
Humidity Sensation Vote (HSV)						
Dry & Very dry	0[a]				1	
Neutral	-0.57	0.19	9.06	0.003*	0.57	(0.39, 0.82)
Humid & Very humid	-0.10	0.22	0.19	0.663	0.91	(0.59, 1.40)
Wind Sensation Vote (WSV)						
No wind	0[a]				1	
Light wind	-1.41	0.18	64.82	<0.001*	0.24	(0.17, 0.34)
Windy	-2.31	0.40	33.82	<0.001*	0.10	(0.05, 0.22)
Shade conditions						
Shaded	0[a]				1	
Exposed	0.50	0.19	6.83	0.009*	1.64	(1.13, 2.38)
Sun Sensation Vote (SSV)						
Dark & Very dark	0[a]				1	
Slightly dark	1.05	0.47	5.03	0.025*	2.86	(1.14, 7.17)
Neutral	1.45	0.45	10.61	0.001*	4.27	(1.78, 10.24)
Slightly bright	0.87	0.49	3.13	0.077	2.39	(0.91, 6.30)
Bright	2.09	0.52	16.12	<0.001*	8.06	(2.91, 22.34)
Very bright	3.75	0.60	38.69	<0.001*	42.32	(13.00, 137.73)
Glare Sensation Vote (GSV)						
Not glaring	0[a]				1	
Slightly glaring	1.22	0.23	28.18	<0.001*	3.39	(2.16, 5.32)
Glaring & Very glaring	1.53	0.33	21.25	<0.001*	4.60	(2.41, 8.81)
Noise Annoyance Vote (NAV)						
Not annoying	0[a]				1	
Slightly annoying	-0.21	0.17	1.53	0.216	0.81	(0.58, 1.13)
Annoying & Very annoying	-0.18	0.27	0.47	0.495	0.83	(0.49, 1.41)
Smelliness						
Not smelly	0[a]				1	
Smelly	-1.40	0.26	28.15	<0.001*	0.25	(0.15, 0.41)
Air Dustiness Vote (ADV)						
Dusty & Very dusty	0[a]				1	

Neither dusty nor clean	0.12	0.25	0.21	0.644	1.12	(0.68, 1.85)
Clean	0.10	0.35	0.08	0.782	1.10	(0.55, 2.19)
Very clean	-0.51	0.56	0.82	0.365	0.60	(0.20, 1.81)
Air Quality Vote (AQV)						
Bad & Very bad	0[a]				1	
Neither poor nor good	0.02	0.32	0.00	0.952	1.02	(0.54, 1.92)
Good	-0.60	0.42	2.08	0.149	0.55	(0.24, 1.24)
Very good	-1.30	0.61	4.47	0.035*	0.27	(0.08, 0.91)
Irritability						
Not irritated	0[a]				1	
Slightly irritated	0.77	0.19	17.21	<0.001*	2.16	(1.50, 3.11)
Irritated & Very irritated	2.09	0.39	28.36	<0.001*	8.09	(3.75, 17.46)
Tiredness						
Not tired	0[a]				1	
Slightly tired	0.48	0.17	7.57	0.006*	1.61	(1.15, 2.27)
Tired & Very tired	-0.10	0.32	0.10	0.755	0.91	(0.48, 1.69)

Note [a] = reference category, * $p < 0.05$, CI = confidence interval for odds ratio (OR)

Table A4 Multivariate ordinal logistic results of thermal tolerance vote (TTV), perceived environmental quality and psychological status. The TTV is divided into five groups (tolerable: 0, slightly difficult to tolerate: +1, fairly difficult to tolerate: +2, very difficult to tolerate: +3, and intolerable: +4). The table shows the unstandardized coefficients (*B*), standard error (*SE*), wald statistics (Wald), p-value (*p*), odds ratio (OR), and 95% confidence intervals (95% CI). An OR value greater than 1 means that people are more likely to feel more intolerable.

Variable	*B*	*SE*	Wald	*p*	OR	95% CI
Location						
Knowledge City	0[a]				1	
Haizhu Wetland Park	1.41	0.16	73.67	<0.001*	4.10	(2.97, 5.66)
Sunny Yongqingfang	0.98	0.14	52.37	<0.001*	2.66	(2.04, 3.48)
Cloudy Yongqingfang	-0.18	0.15	1.49	0.222	0.83	(0.62, 1.12)
Gender						
Male	0[a]				1	
Female	0.10	0.10	0.85	0.356	1.10	(0.90, 1.35)
Humidity Sensation Vote (HSV)						
Dry & Very dry	0[a]				1	
Neutral	-0.40	0.13	10.16	0.001*	0.67	(0.52, 0.86)
Humid & Very humid	-0.23	0.15	2.43	0.119	0.79	(0.60, 1.06)
Wind Sensation Vote (WSV)						
No wind	0[a]				1	
Light wind	-1.27	0.11	139.24	<0.001*	0.28	(0.23, 0.35)
Windy	-2.30	0.24	91.23	<0.001*	0.10	(0.06, 0.16)
Shade conditions						
Shaded	0[a]				1	
Exposed	0.42	0.13	10.48	0.001*	1.52	(1.18, 1.95)
Sun Sensation Vote (SSV)						
Dark & Very dark	0[a]				1	
Slightly dark	1.08	0.25	19.07	<0.001*	2.94	(1.81, 4.77)
Neutral	1.38	0.23	35.83	<0.001*	3.99	(2.54, 6.29)
Slightly bright	1.42	0.26	29.06	<0.001*	4.13	(2.46, 6.91)
Bright	2.16	0.30	53.57	<0.001*	8.70	(4.87, 15.52)
Very bright	3.88	0.36	113.32	<0.001*	48.28	(23.65, 98.58)
Glare Sensation Vote (GSV)						
Not glaring	0[a]				1	
Slightly glaring	1.25	0.16	64.25	<0.001*	3.49	(2.57, 4.73)
Glaring & Very glaring	1.92	0.24	64.61	<0.001*	6.81	(4.26, 10.86)
Noise Annoyance Vote (NAV)						
Not annoying	0[a]				1	
Slightly annoying	0.29	0.11	6.88	0.009*	1.33	(1.08, 1.66)
Annoying & Very annoying	0.42	0.17	5.74	0.017*	1.51	(1.08, 2.13)
Smelliness						
Not smelly	0[a]				1	
Smelly	-0.49	0.15	10.59	0.001*	0.61	(0.45, 0.82)
Air Dustiness Vote (ADV)						

Dusty & Very dusty	0[a]				1	
Neither dusty nor clean	0.14	0.17	0.69	0.406	1.15	(0.82, 1.62)
Clean	0.40	0.23	2.98	0.084	1.49	(0.95, 2.35)
Very clean	0.05	0.34	0.02	0.893	1.05	(0.53, 2.05)
Air Quality Vote (AQV)						
Bad & Very bad	0[a]				1	
Neither poor nor good	0.12	0.21	0.32	0.573	1.13	(0.74, 1.72)
Good	-0.40	0.27	2.15	0.143	0.67	(0.39, 1.14)
Very good	-1.25	0.38	10.69	0.001*	0.29	(0.14, 0.61)
Irritability						
Not irritated	0[a]				1	
Slightly irritated	0.44	0.13	12.05	<0.001*	1.56	(1.21, 2.00)
Irritated & Very irritated	1.43	0.25	33.43	<0.001*	4.18	(2.58, 6.80)
Tiredness						
Not tired	0[a]				1	
Slightly tired	0.25	0.12	4.87	0.027*	1.29	(1.03, 1.62)
Tired & Very tired	0.59	0.19	9.59	0.002*	1.81	(1.24, 2.64)

Note [a] = reference category, * $p < 0.05$, CI = confidence interval for odds ratio (OR)

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Supplementary material – survey questionnaire

Baseline survey before the thermal walk experiment

Participant ID: _____

Gender: ☐1 Male ☐2 Female

Age: ___ **Height:**____cm **Weight:**____kg

Place of origin: Province _____ City _____

How long have you been in Guangzhou?

☐1 <3 months ☐2 3-12 months ☐3 1-3 years ☐4 3-10 years ☐5 >10 years

Tick all appropriate boxes to indicate the respondent's current attire. (*Dress = Short/Long sleeved shirt/singlet top + short/long skirt)

Upper body	☐ Singlet top	☐ Short sleeved shirt	☐ Long sleeved shirt	☐ Topless
	☐ Hat	☐ Headscarf	☐ Jacket	☐ Sunglasses
Upper body (Fabric)	☐ Cotton	☐Synthetic(polyester/chiffon)	☐Others _____	
Lower body	☐ Shorts	☐ Yoga pants/tights	☐ Jeans	☐ Other long pants
	☐ Short skirt	☐ Long skirt		
Footwear	☐ Shoes	☐ Socks	☐ Sandals	☐ Slippers

Table S1 Clothing insulation values of individual clothing garments derived from ISO 9920 (2007). The clothing insulation value of underwear (men's brief or panties and bras) is 0.04 (ISO 9920, 2007), which is added to the overall clothing insulation values of survey respondents.

Garment description	Clothing insulation (clo)
Singlet top	0.04
Short-sleeved shirt	0.15
Long-sleeved shirt	0.25
Jacket (light summer jacket)	0.25
Hat (cap)	0.01
Headscarf*	0.27
Shorts	0.06
Yoga pants/tights	0.25
Jeans	0.25
Other long pants	0.25
Short skirt	0.15
Long skirt	0.25
Shoes	0.04
Socks	0.02
Sandals	0.02
Slippers	0.02

* Headscarf = 0.26 (Long-sleeve turtleneck (thin)) + 0.01 (cap)

Survey questions for the thermal walk experiments

Date: _____ Time:____:____ Participant ID:___

Survey location:_____

Questions for assessing thermal environments (adapted from ISO 10551 (2019))

Q1: How are you feeling now? (adapted from Fang et al. (2019))

Cold -3	Cool -2	Slightly cool -1	Neutral 0	Slightly warm +1	Warm +2	Hot +3	Very hot +4	Extremely hot +5

Q2: How do you feel about the humidity now? (adapted from Lam et al. (2021a))

☐$_{-2}$ Very dry ☐$_{-1}$ Dry ☐$_{0}$ Neutral ☐$_{+1}$ Humid ☐$_{+2}$ Very humid

Q3: How do you feel about the wind now? (adapted from Lam et al. (2021a))

☐$_{-2}$ No wind ☐$_{-1}$ Little wind ☐$_{0}$ Neutral ☐$_{+1}$ Windy ☐$_{+2}$ Too much wind

Q4: Do you find this thermal environment…?

☐$_{0}$ Comfortable ☐$_{1}$ Slightly uncomfortable ☐$_{2}$ Uncomfortable
☐$_{3}$ Very uncomfortable ☐$_{4}$ Extremely uncomfortable

Q5: Please state how you would prefer to be now.

☐$_{-1}$ Cooler ☐$_{0}$ No change ☐$_{1}$ Warmer

Q6: How do you judge this thermal environment on a personal level?

☐$_{0}$ Acceptable ☐$_{1}$ Unacceptable

Q7: Do you find this thermal environment…?

☐$_{0}$ Tolerable ☐$_{1}$ Slightly difficult to tolerate ☐$_{2}$ Fairly difficult to tolerate
☐$_{3}$ Very difficult to tolerate ☐$_{4}$ Intolerable

Questions for assessing the visual environments (adapted from ISO 10551 (2019))

Q8: Are you under shade now?

☐$_{1}$**Yes** ☐$_{2}$**No**

Q9: How are you feeling about the sunlight now?

Very dark -3	Dark -2	Slightly dark -1	Neutral 0	Slightly bright 1	Bright 2	Very bright 3

Q10: For the current spot you are at now, would you like it to be…?

☐$_{-1}$ Darker ☐$_{0}$ No change ☐$_{1}$ Brighter

Q11: How do you rate about the magnitude of the glare in this environment now? (adapted from Iwata and Tokura (1998))

☐$_0$ Not glaring ☐$_1$ Slightly glaring ☐$_2$ Glaring ☐$_3$ Very glaring

<u>Questions for assessing acoustic environments (adapted from ISO 10551 (2019))</u>

Q12: How do you feel about the noise in this environment now?

☐$_0$ Not annoying ☐$_1$ Slightly annoying ☐$_2$ Annoying ☐$_3$ Very annoying

Q13: To what extent do you currently hear the following types of sound? (Please select up to two sounds) (adapted from Jeon and Jo (2020))

☐$_0$ Traffic ☐$_1$ Construction ☐$_2$ Human ☐$_3$ Animal (e.g., birdsong, dog barking)
☐$_4$ Wind, leaf, water ☐$_5$ Music ☐$_6$ Other (please specify):_____

Q14: Please state how you would prefer to be now.

☐$_0$ No change ☐$_1$ Slightly quieter ☐$_2$ Quieter ☐$_3$ Much quieter

<u>Questions for assessing air quality environments</u>

Q15: Do you find the air to be…? (adapted from ISO 10551 (2019))

☐$_0$ Not smelly ☐$_1$ Slightly smelly ☐$_2$ Smelly ☐$_3$ Very smelly

Q16: What is your opinion of the air quality in relation to dust at the moment? (adapted from Pantavou et al. (2017))

☐$_{-2}$ Very dusty ☐$_{-1}$ Dusty ☐$_2$ Neither dusty nor clean ☐$_3$ Clean ☐$_4$ Very clean

Q17: What is your opinion of the overall air quality at the moment? (adapted from Pantavou et al. (2017))

☐$_{-2}$ Very poor ☐$_{-1}$ Poor ☐$_2$ Neither poor nor good ☐$_3$ Good ☐$_4$ Very good

<u>Questions to assess participant's psychological conditions (adapted from Lam et al. (2021b))</u>

Q18: How is your current mental state?

☐$_0$ Not irritated ☐$_1$ Slightly irritated ☐$_2$ Irritated ☐$_3$ Very irritated

Q19: How is your current perceived fatigue state?

☐$_0$ Not tired ☐$_{1`}$ Slightly tired ☐$_2$ Tired ☐$_3$ Very tired

Supplementary tables and figures

Table S2 Summary table of the chi-square test of association, showing the association between thermal comfort indicators (TSV, TCV) and visual and acoustic comfort indicators (SSV and GSV over various horizontal illuminance ranges, and NAV over various PET ranges).

TSV				
Variable	χ^2	*df*	*p*	ϕ_c
SSV (1-2000 lx)	15.304	6	0.018	0.110
SSV (2000-20000 lx)	19.248	6	0.004	0.093
SSV (20000–110000 lx)	228.508	6	<0.001	0.431
GSV (1-2000 lx)	13.098	3	0.004	0.144
GSV (2000-20000 lx)	60.954	6	<0.001	0.166
GSV (20000–110000 lx)	311.630	9	<0.001	0.411
NAV (29.1 °C–36.3 °C)	8.756	6	0.188	0.057
NAV (36.3 °C–53.6 °C)	19.705	6	0.003	0.100
TCV				
SSV (1-2000 lx)	39.025	8	<0.001	0.176
SSV (2000-20000 lx)	47.257	8	<0.001	0.146
SSV (20000–110000 lx)	226.331	8	<0.001	0.429
GSV (1-2000 lx)	16.226	4	0.003	0.160
GSV (2000-20000 lx)	109.826	8	<0.001	0.223
GSV (20000–110000 lx)	411.471	12	<0.001	0.472
NAV (29.1 °C–36.3 °C)	53.076	8	<0.001	0.140
NAV (36.3 °C–53.6 °C)	72.616	8	<0.001	0.191

Note for abbreviation: TSV: thermal sensation vote, TCV: thermal comfort vote, SSV: sun sensation vote, GSV: glare sensation vote, NAV: noise annoyance vote.

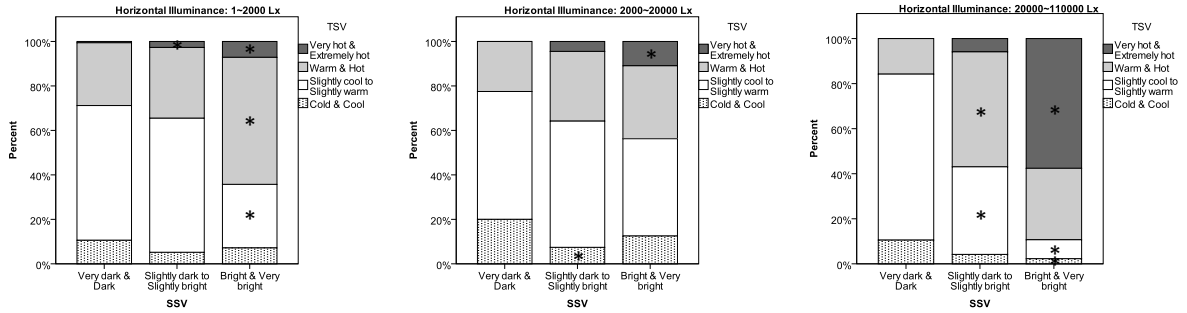


Fig. S1 The percentage of thermal sensation votes (TSV) for respondents with different sun sensation votes (SSV) over different horizontal illuminance ranges. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

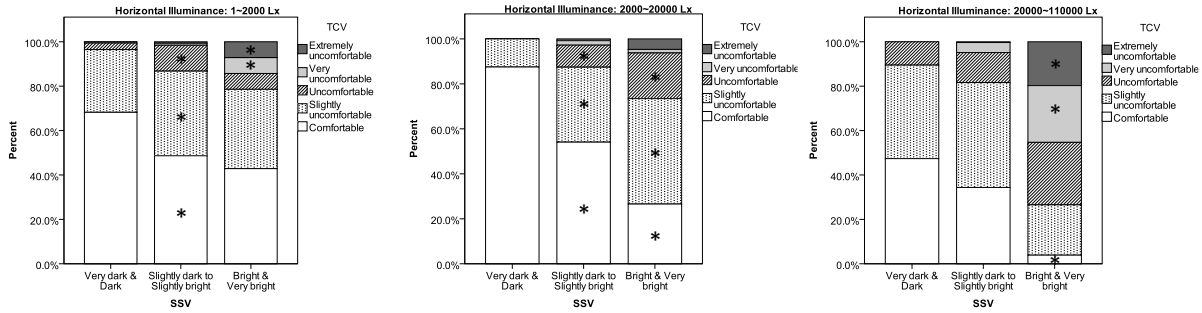


Fig. S2 The percentage of thermal comfort votes (TCV) for respondents with different sun sensation votes (SSV) over different horizontal illuminance ranges. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

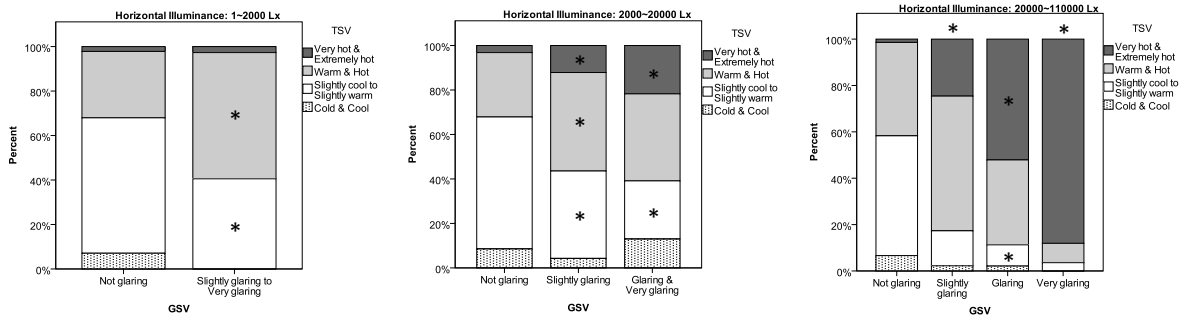


Fig. S3 The percentage of thermal sensation votes (TSV) for respondents with different glare sensation votes (GSV) over different horizontal illuminance ranges. When horizontal illuminance was 1–2000 lx and 2000–2000 lx, some GSV groups were merged due to limited sample size in a certain GSV group. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

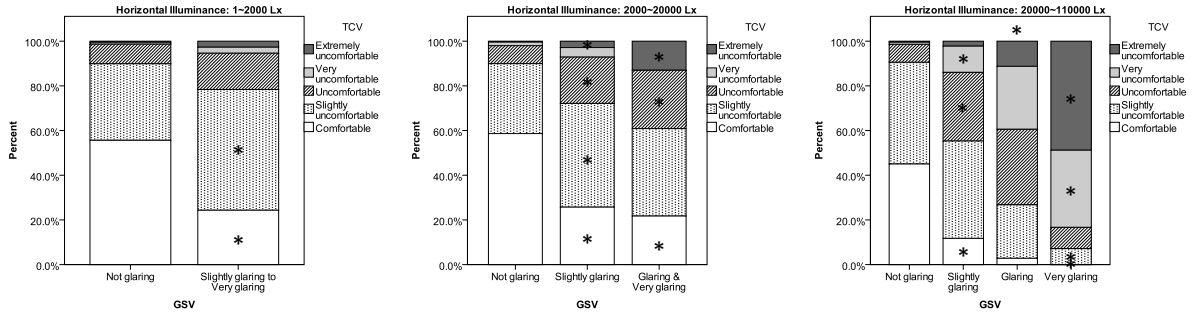


Fig. S4 The percentage of thermal comfort votes (TCV) for respondents with different glare sensation votes (GSV) over different horizontal illuminance ranges. When horizontal illuminance was 1–2000 lx and 2000–2000 lx, some GSV groups were merged due to limited sample size in a certain GSV group. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

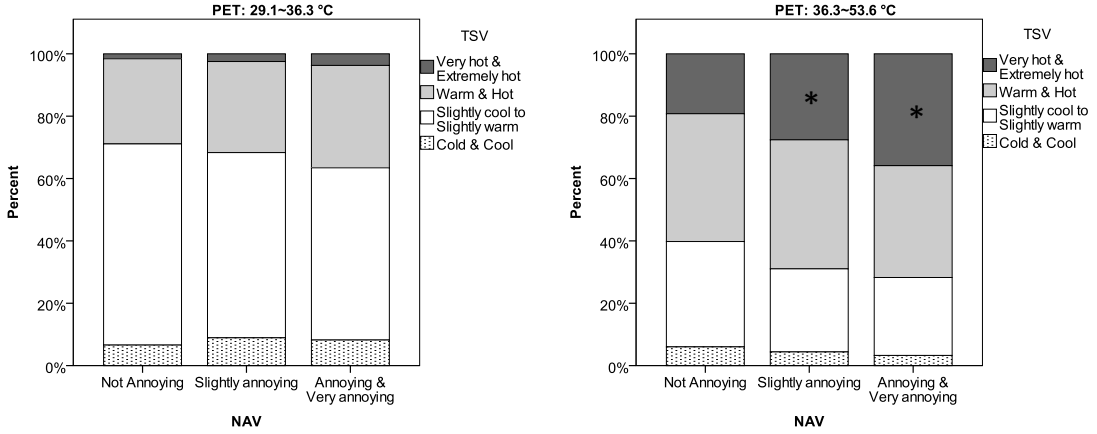


Fig. S5 The percentage of thermal sensation votes (TSV) for respondents with different noise annoyance votes (NAV) over different Physiological Equivalent Temperature (PET) range**s**. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

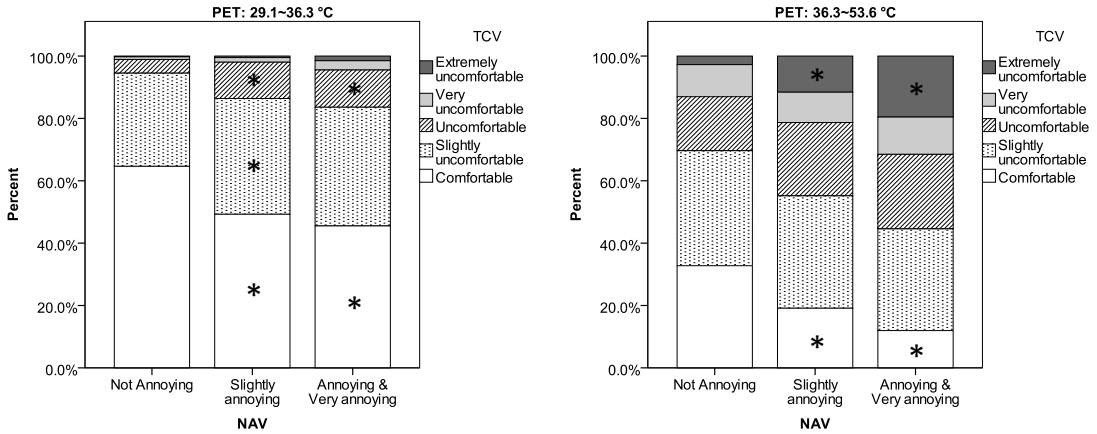


Fig. S6 The percentage of thermal comfort votes (TCV) for respondents with different noise annoyance votes (NAV) over different Physiological Equivalent Temperature (PET) ranges. The asterisk (*) indicates the groups that are significantly different in proportion to the reference group on the left.

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